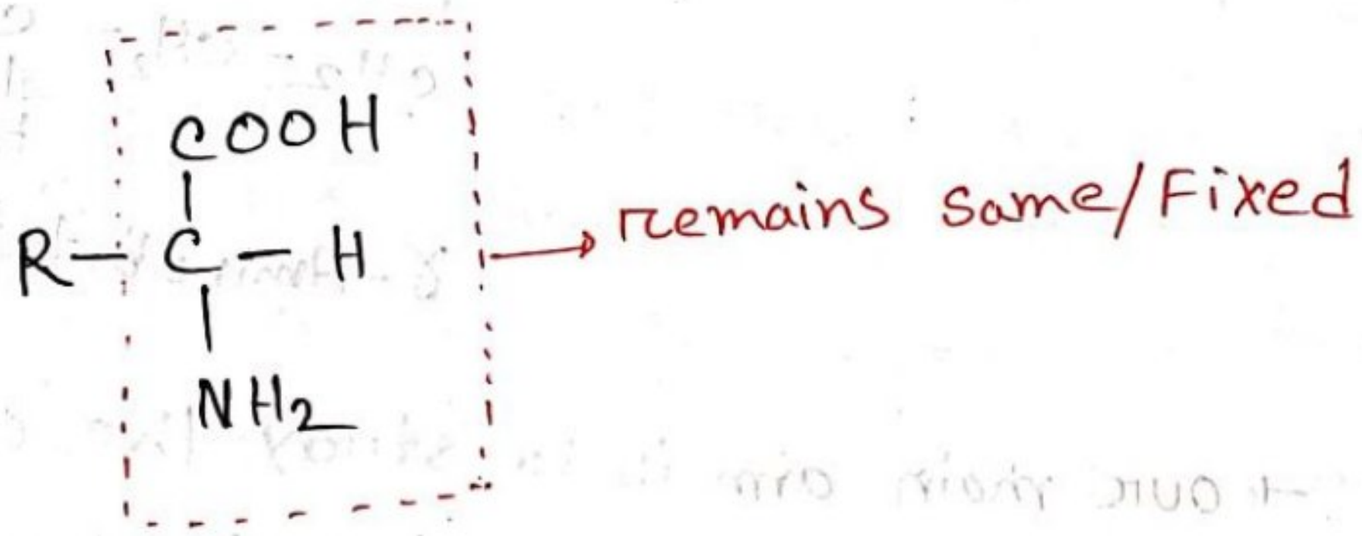
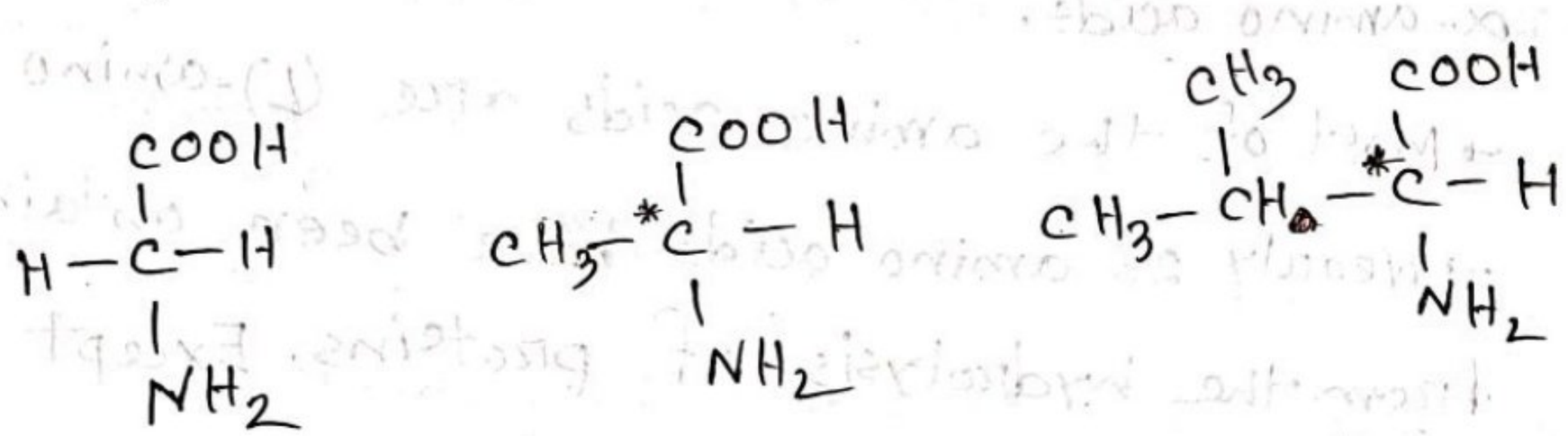
ALVEE
Chemistry-48

Amino acids and proteins

Amino acids: Amino acids are the compounds having one or more amino ($-NH_2$) group and carboxylic acid ($-COOH$) group in the same molecule. Except Glycine(G), all other amino acids consist an organic R group. So the general structure of an amino acid is -



Examples: Glycine, Alanine, Valine

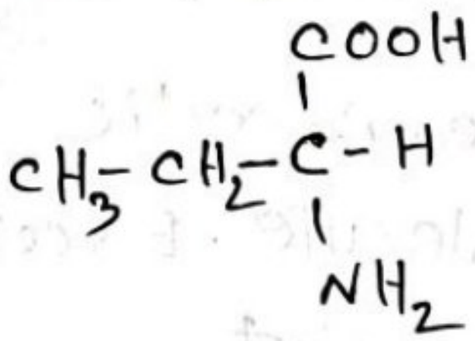


classification of amino acids:

① on the basis of the position of the carboxylic acid group, amino acids are classified in the followings -

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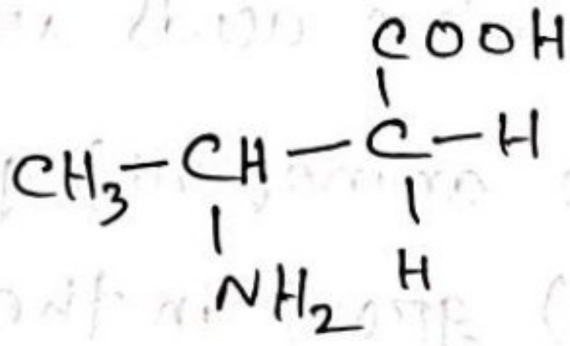
① α-amino acid



α-Aminobutanoic acid

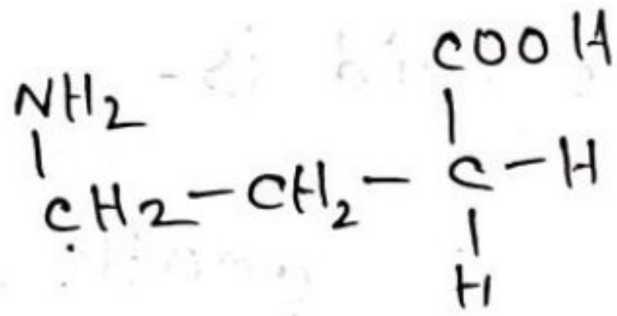
② Depe... type...

⑪ β-amino acid



β-Aminobutanoic acid

⑪ γ-amino acid



γ-Aminobutanoic acid.

→ our main aim is to study the chemistry of proteins which on hydrolysis give exclusively α-amino acids.

→ Most of the amino acids are (L)-amino acid.

→ Nearly 25 amino acids have been obtained from the hydrolysis of proteins. Except two (proline and hydroxyproline) all are amino acids, while the exceptional two are imino acids.

(a) 23 α amino acid

(b) 2 α imino acid (−NH− or =NH)

Proteins $\xrightarrow[\text{Acid/base catalyzed hydrolysis}]{\text{Enzyme or}}$ α-amino acid

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② Depending on availability amino acids are two types -

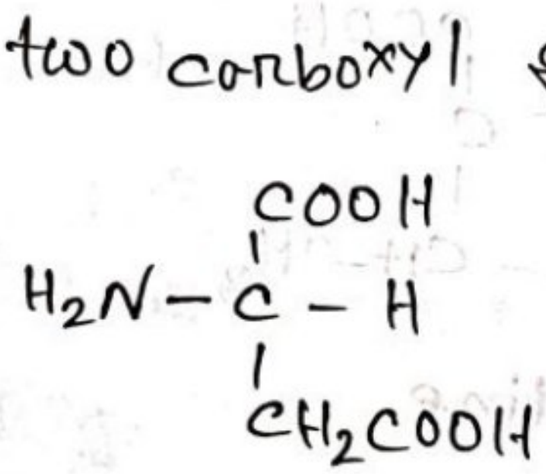
① Non-essential amino acids (also called dispensable amino acid). These amino acids are synthesised in human body. 11 out of 20 amino acids are known to be non-essential. Example: Arginine, alanine, aspartic acid, glutamin, glutamic acid proline, glycine, serine, tyrosine, asparagine, cystenine.

② Essential amino acids cannot be synthesized by the body and are required from food supplements. There are 9 essential amino acids that include leucine, isoleucine, histidine, lysine, methionine, threonine, phenylalanine, tryptophan and valine.

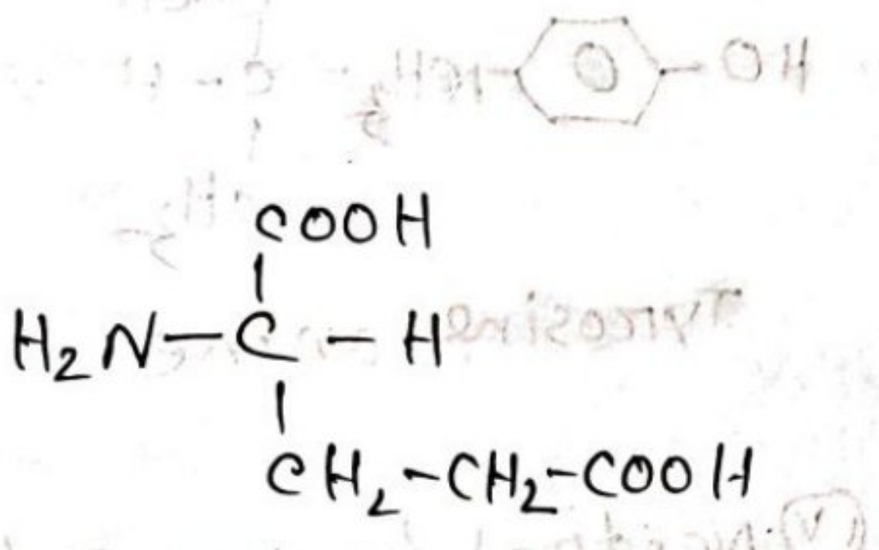
* Human body can utilize completely only the natural L-isomer of the essential amino acids.

③ According to the properties of the side chain amino acids are classified in 3 types-

① Acidic amino acid: contains one amino and two carboxyl group.

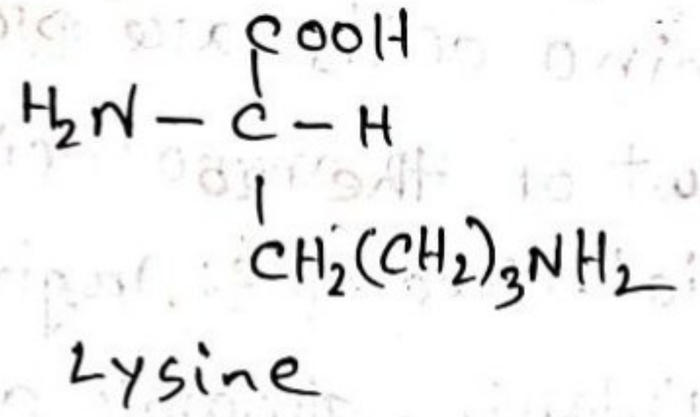


Aspartic acid

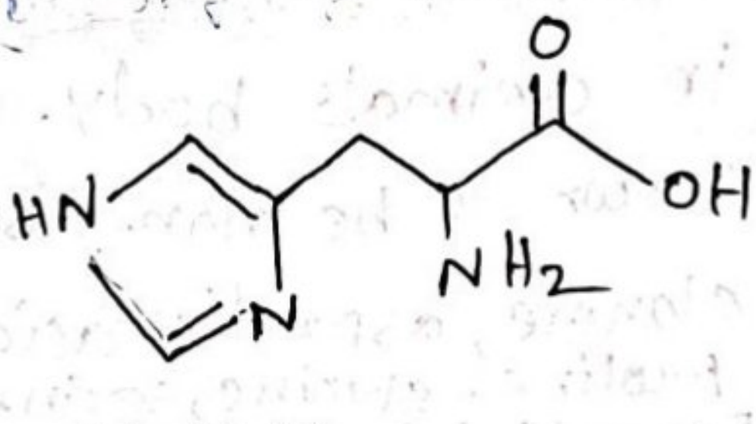


α-aminoglutamic acid

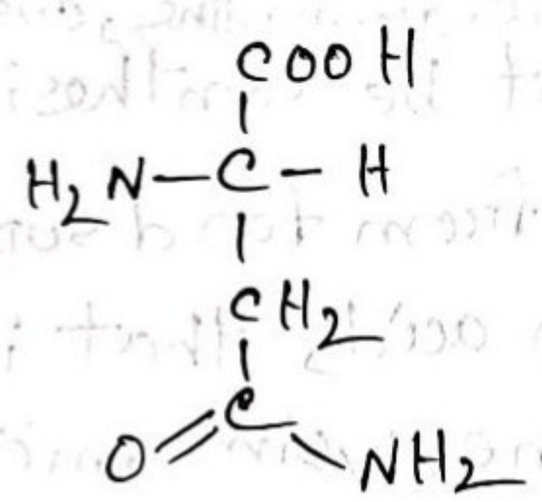
(ii) Basic amino acid: This contain a basic side ch
(have amino group in side chain). There are 4 basic amino acid.
These are- Lysine, Arginine, Histidine, ornithine.



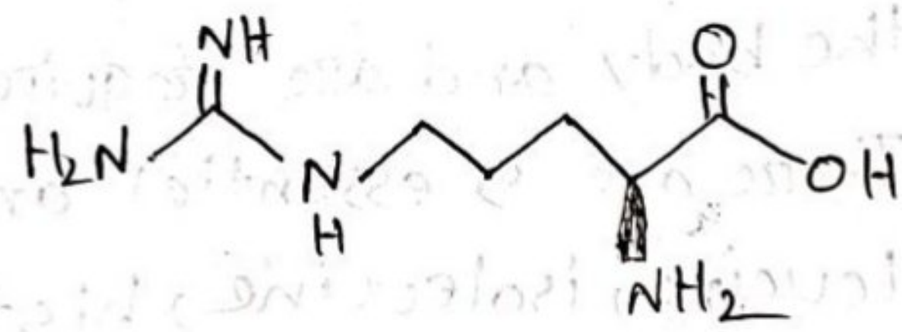
Lysine



Histidine



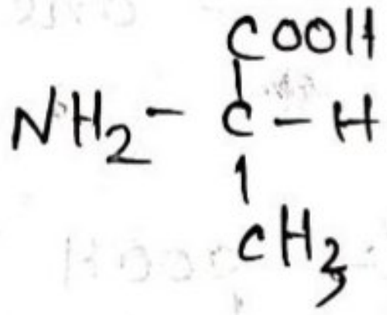
Asparagine



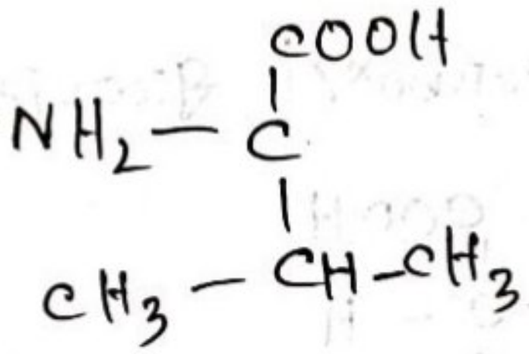
Arginine

(since amide group is neutral asparagine also considered as neutral Polar amino acid)

(iii) Neutral non polar amino acid:
If the side chain is neutral then the amino acid is called neutral amino acid. Because the amino and carboxilic group nectralizes each other. Ex-



Alanine

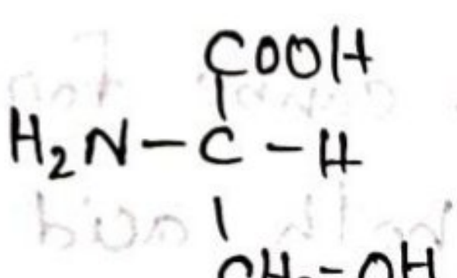


valine

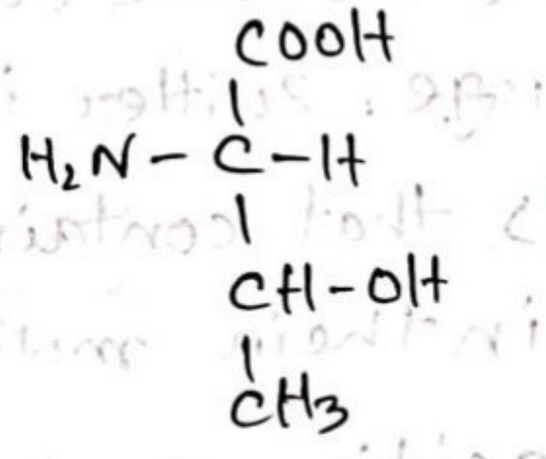
(iv) Neutral/uncharged polar amino acid:
If the side chain is neutral but polar then its called neutral polar amino acid.

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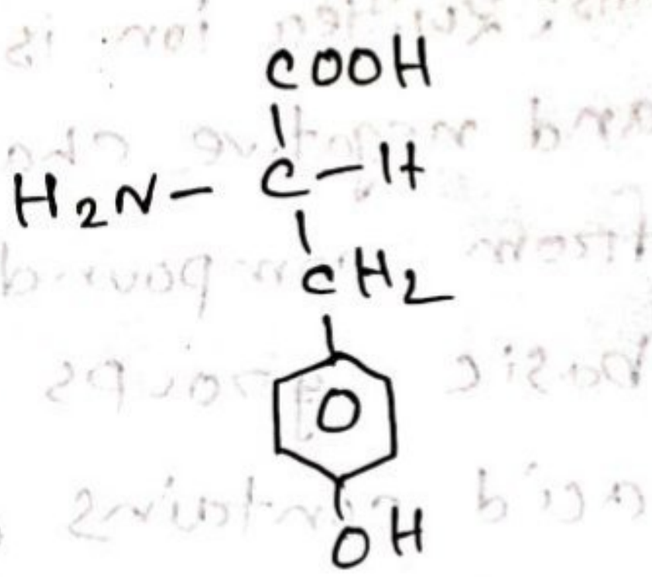
Ex.



Serine

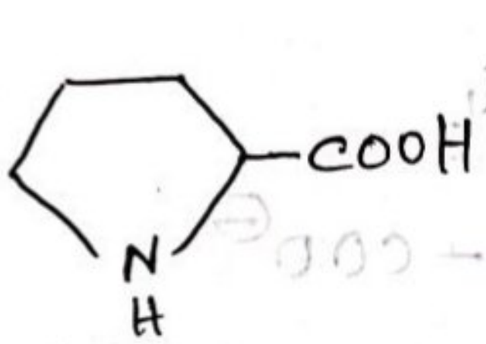


Threonine

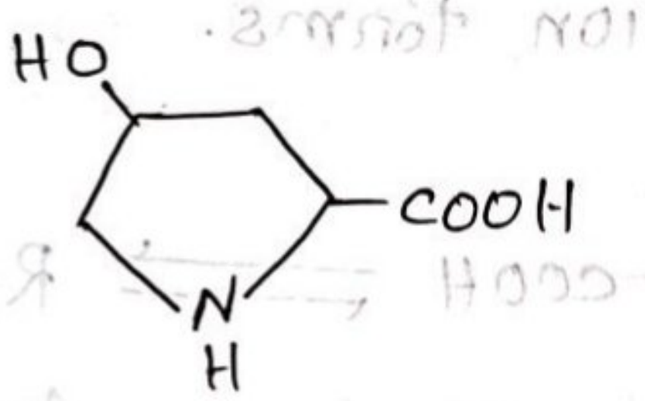


Tyrosine

The structure of two imino acids are—



Proline



Hydroxyproline

Physical properties of amino acids;

① Colourless crystalline solid.

② Generally soluble in water but sparingly soluble in organic solvent

③ Most melt with decomposition, but sublimation is possible with a number of amino acids.

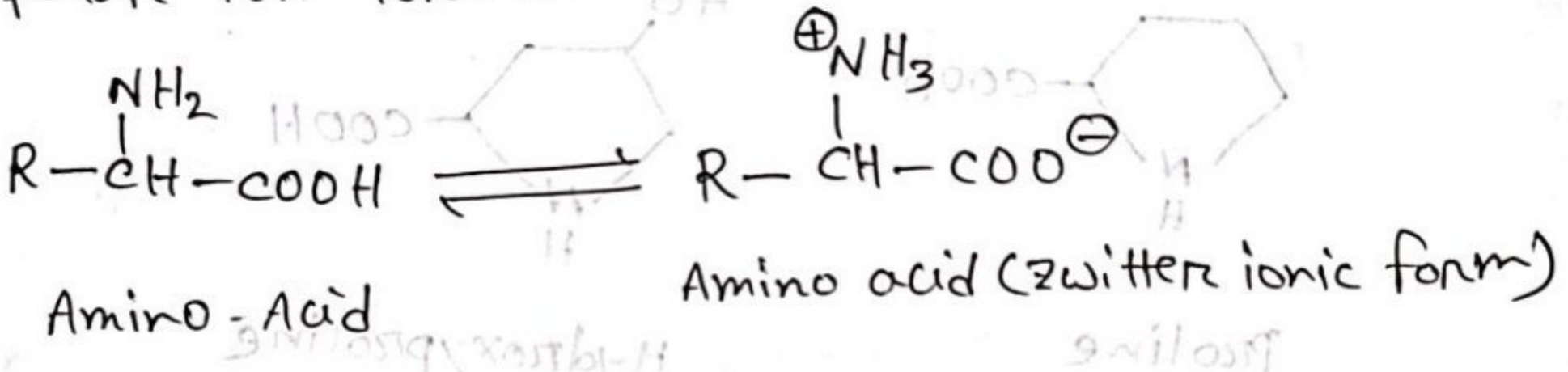
④ Aqueous solution of amino acids have very high dielectric constant.

⑤ Acidity and basicity constants are ridiculously low for -COOH and $-NH_2$ groups.

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#"Amino acid forms zwitter ion" explain-

Ans: Zwitter ion is a molecule contains both positive and negative charge. Zwitter ions canbe formed from compounds that contain both acid and basic groups in their molecule. Since amino acid contains acidic carboxylic group (-COOH) and basic aming group ($-NH_2$), carboxilic acid group protonates basic amino group thus dipolar ion forms.



Amino-Acid

Amino acid (zwitter ionic form)

Extra

The presence of dipolar ion structure of amino acid is evidenced by the following points.

(i) High m.p., solubility in water, insolubility in ether or other organic solvent

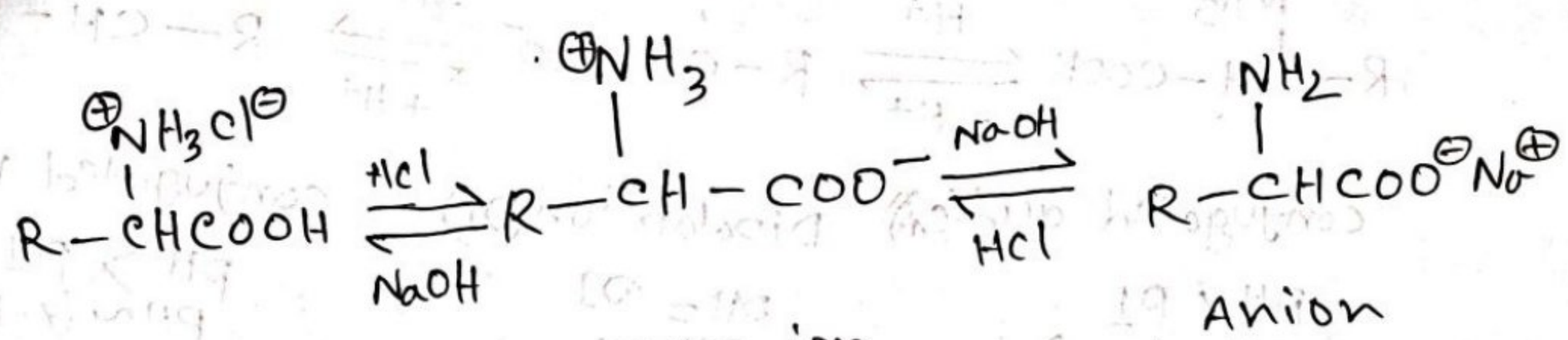
(ii) solution having high dielectric constant

(iii) The spectroscopic studies show the absence of the vibration frequencies for the free carboxyl and amino groups.

(iv) It explains differen electrolytic behaviour in diff. media, in an acidic solution an amino acid behaves as a protonated derivative hence migrate to the cathode; while in alkaline medium the same amino acid behaves as anion derivative

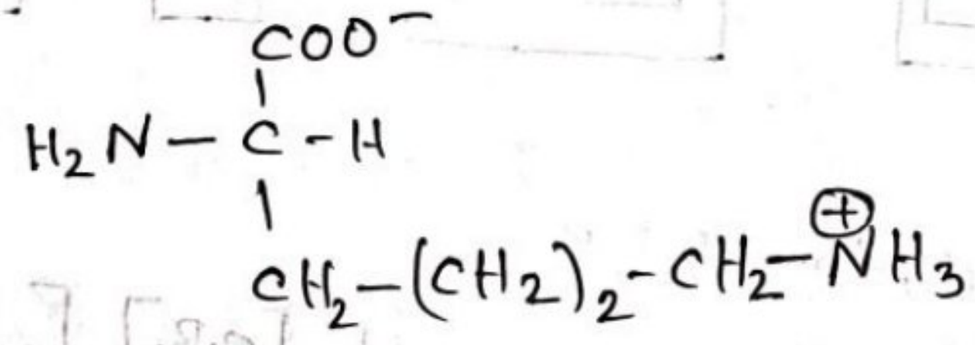
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and hence migrates to the anode.



Note Extra

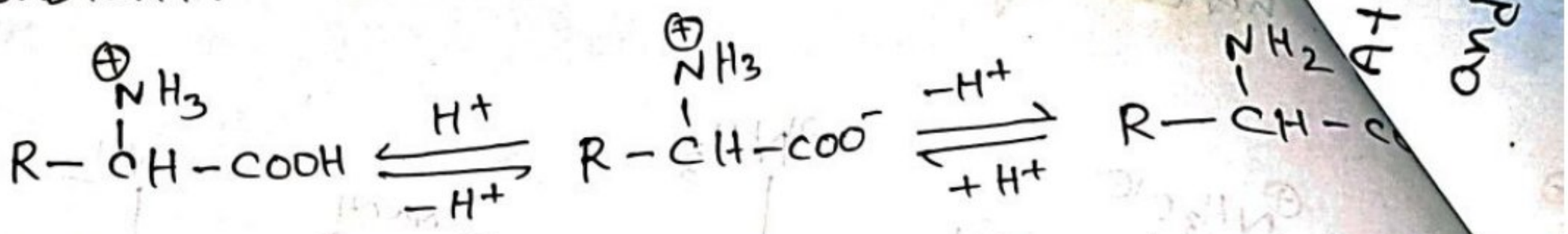
Such- zwitter ionic form of basic amino acid



Protonation occurs at side chain due to direct connection of electron releasing group with the basic amino group in the side chain. Electron releasing group increases its basicity.

Isoelectric point:- The pH at which there is no net charge on the amino acid molecule and which thus does not migrate to any electrode under the influence of electric current is known as isoelectric point. The isoelectric point of each amino acid is

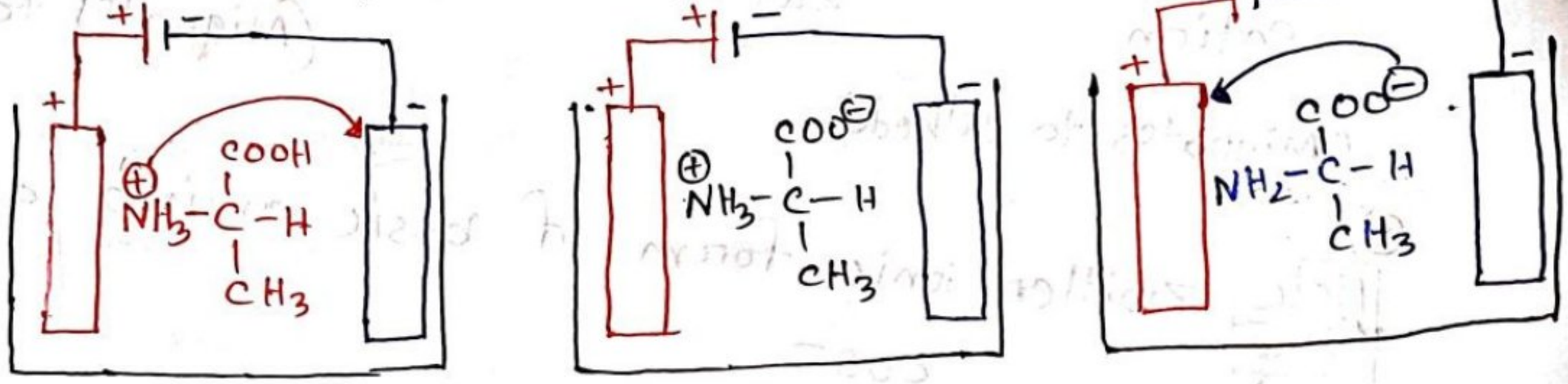
constant.



conjugated acid(CA) | Dipolar ion(DI) | conjugated base

pH < PI | pH = PI | pH > PI

pH ~ (0-7) | pH = ? | pH ~ (7-14)



How to calculate PI?

Ans: To calculate the isoelectric point of an amino acid, representing the iso-electric amino acid as $H_3\overset{+}{N}-Z-CO_2^-$, then we have the following equilibria

$$H_3\overset{+}{N}-Z-CO_2H \rightleftharpoons H_3\overset{+}{N}-Z-CO_2^- + H^+$$

CA D.I

$$H_3\overset{+}{N}-Z-CO_2^- \rightleftharpoons H_2N-Z-CO_2^- + H^+$$

D.I. C.B

$$K_1 = \frac{[D.I][H^+]}{[CA]} \qquad K_2 = \frac{[C.B][H^+]}{[D.I]}$$

$$\Rightarrow [CA] = \frac{[DI][H^+]}{K_1} \qquad \Rightarrow [CB] = \frac{K_2[DI]}{[H^+]}$$

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At the isoelectric point (pHi), [D.I] is a maximum and since the net charge is zero,

$$\therefore [CA] = [CB]$$

$$\Rightarrow \frac{[D\cdot I]\,[H_i^+]}{K_1} = \frac{K_2\,[DI]}{[H_i^+]}$$ {Let H_i^+ for isoelectric point}

$$\Rightarrow [H_i^+]^2 = K_1 K_2$$

$$\Rightarrow -\log [H_i^+]^2 = -\log K_1 K_2$$

$$\Rightarrow 2\,pH_i = pK_1 + pK_2$$

$$\Rightarrow pH_i = \frac{pK_1 + pK_2}{2}$$

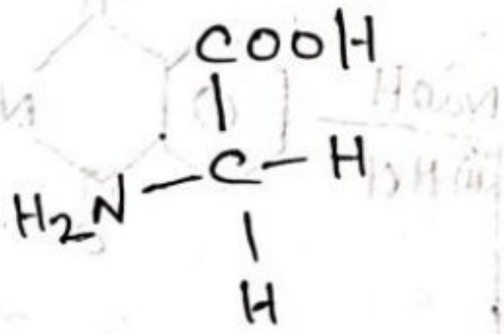
For example,

Two pKa value of glycine is.

$pK_1 = 2.4 \qquad pK_2 = 9.6$

$$pI = \frac{2.4 + 9.6}{2}$$

$$= 6.0$$



Glycine

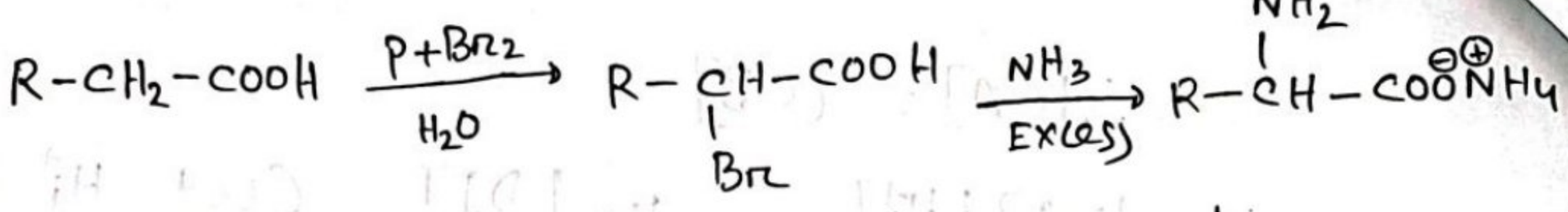
Extra

Iso electric point of neutral amino acid is 5.5 – 6.3.
Isoelectric point of acidic amino acid is 3.0 – 3.5
Isoelectric point of basic amino acid is 7.6 – 10.8

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Synthesis of amino acid

① Direct amination of α-halo acids:

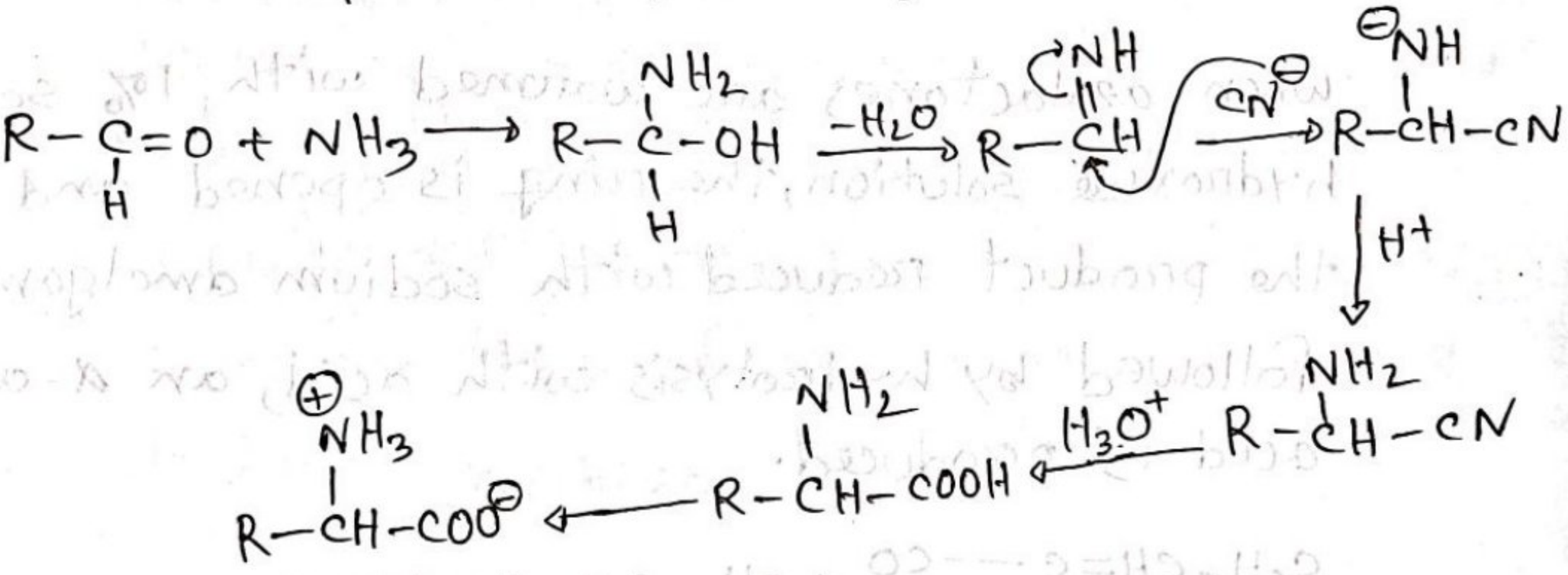


Glycine, alanine, serine, threonine, valine, leucine have been prep. by this method. since this method requires large amount of ammonia and also gives side products like $NH(H_2COOH)_2$, $N(CH_2COOH)_3$; it is replaced by Gabriel pthalamide synthesis.

② Gabriel pthalamide synthesis:

(iii) Strecker synthesis: Aldehyde or ketone is treated with a mixture ammonium chloride and potassium cyanide. The product upon hydrolysis yields the corresponding amino acid. The reaction is initiated by the nucleophilic addition of ammonia to aldehyde. Elemination of water followed by the addition of cyanide to the aladimine yields α-amino nitril which on hydrolysis gives α-amino acid

$$NH_4Cl + KCN \rightleftharpoons NH_3 + HCN + KCl$$



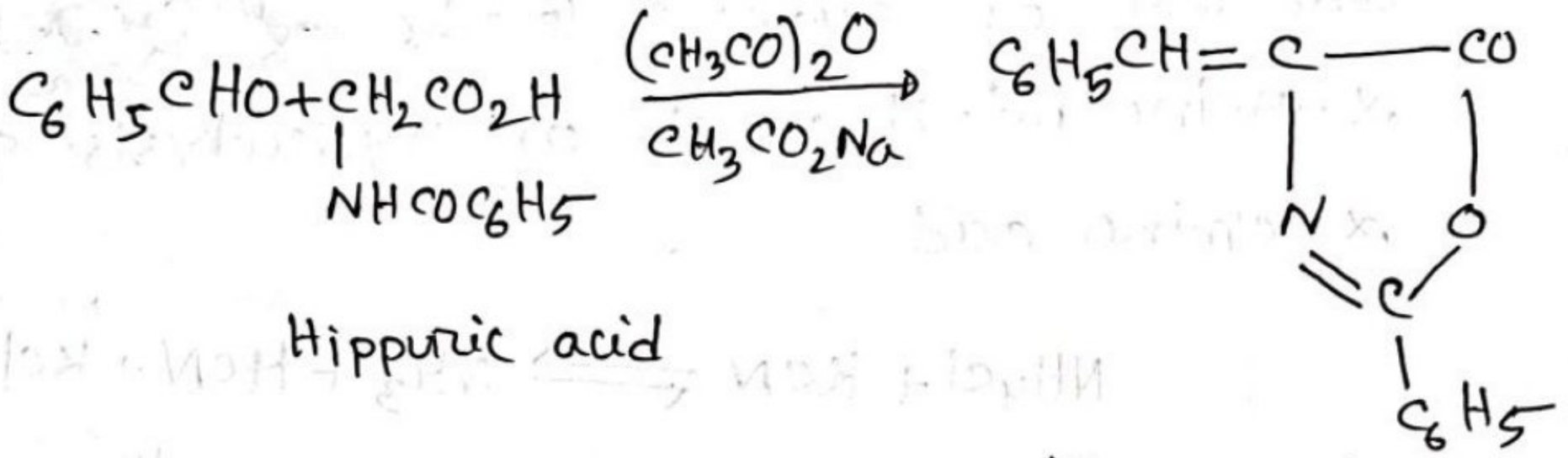
Extra The method is useful for preparing the following amino acids: glycine, valine, alanine, serine, methionine glutamic acid, leucine, isoleucine, norleucine, Phenylalanine

The mechanism of the strecker synthesis is uncertain but the one given above is very highy favored. Optically active α amino acid have been prepared when the reaction carried out in the presence of optically active base (e.g. an alkaloid).

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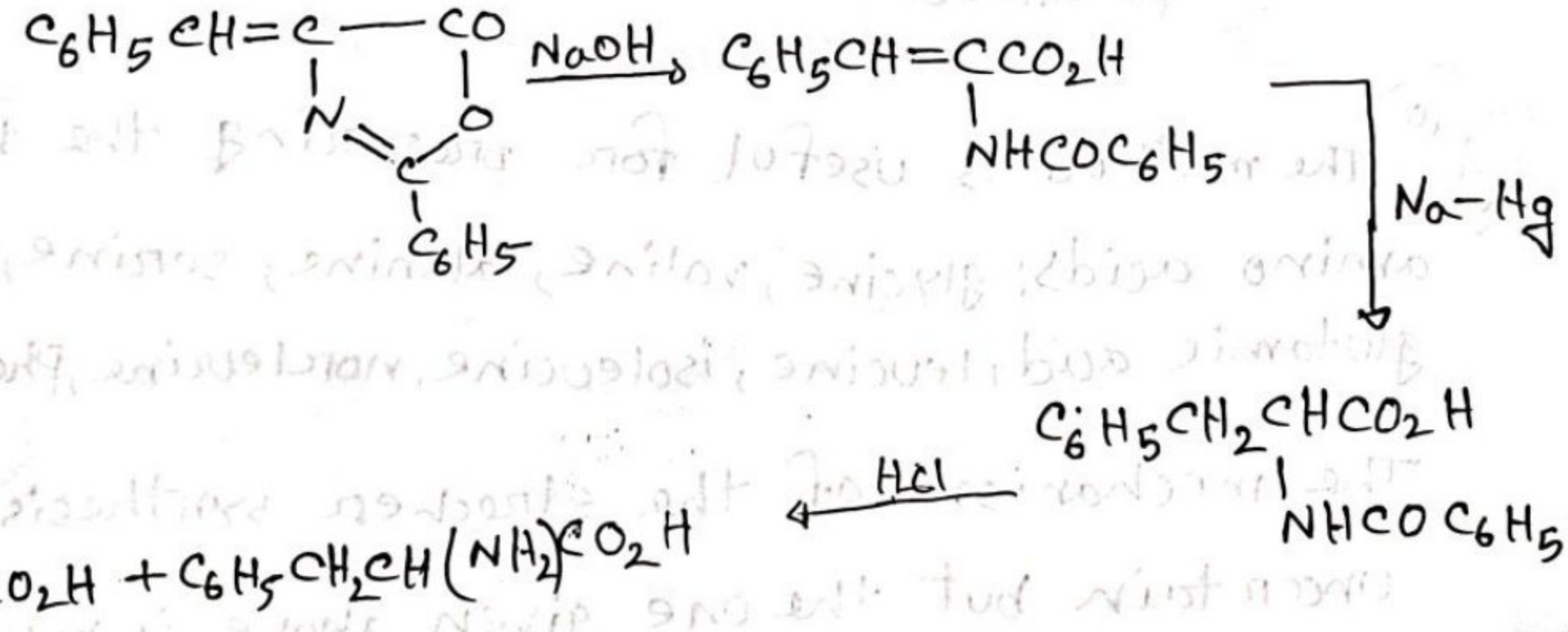
(iv) The Azlactone synthesis (Erlenmeyer synthesis)

Azalpacaltone are usually prepared by heating an aromatic aldehyde with hippuric acid (benzoylglycine) in presence of acetic anhydride and sodium acetate, e.g. benzaldehyde forms benzoyl-α-aminocinnamic azlactone (4-benzylidene-2-phenyloxazol-5-one)



Hippuric acid

When azlactones are warmed with 1% sodium hydroxide solution, the ring is opened and if the product reduced with sodium amalgam followed by hydrolysis with acid, an α-amino acid is produced.



Extra

→ This azlactone synthesis offers a convinient method of preparing phenylalanine, tyrosine, thyroxine, tryptopan.

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→ Aceturic acid ($CH_3\overset{O}{\overset{\|}{C}}NHCH_3CO_2H$) may also be used instead of hippuric acid. Furthermore, it has been found that aliphatic aldehydes may condense with hippuric acid to form azlacetone if lead acetate is used instead of sodium acetate.

Chemical properties of amino acids

The chemical properties of amino acids are generally those of the amino group and the carboxyl group, although some other properties which are due to the presence of amino and carboxyl groups in the same molecule are also responded by the amino acids. The chemical reactions of amino acids can be catagorize in three-

Ⓐ Reactions due to amino group.

Ⓑ Reactions due to carboxyl group.

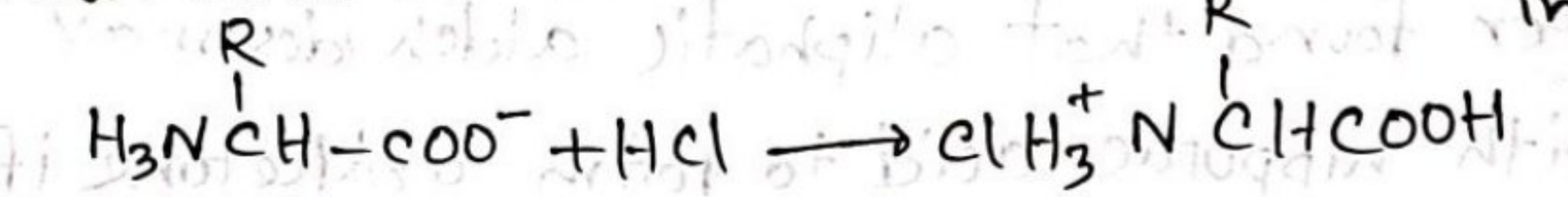
Ⓒ Reactions due to both amino and carboxyl group

Alvee Sahriar Noor Soyeed
Chemistry-48
Date-10/6/2022
5 days needed to make this shit!I mean sheet.

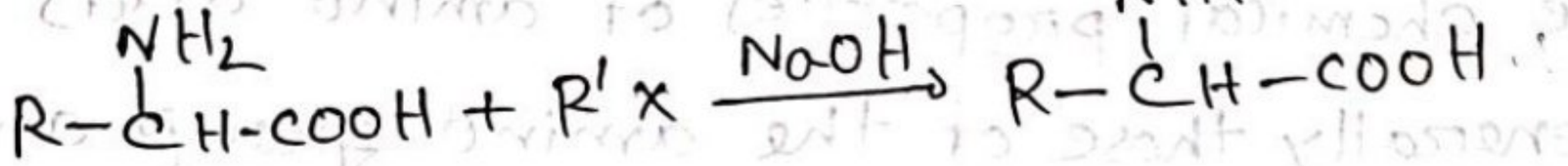
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(A) Reactions due to amino group

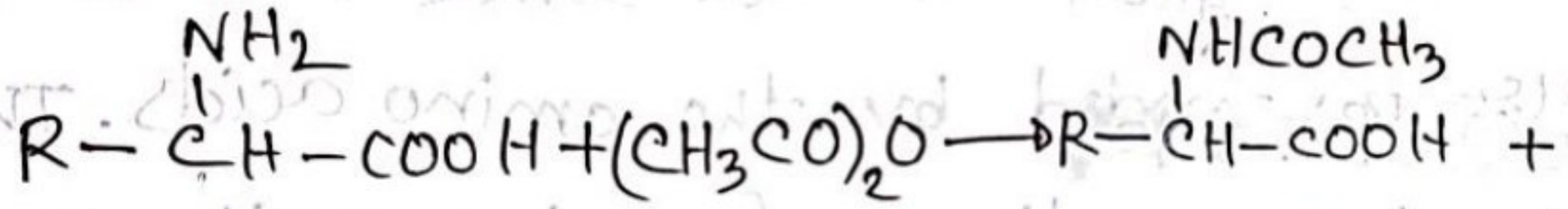
① Action with mineral acids: Forms salt (sparingly soluble in H_2O)



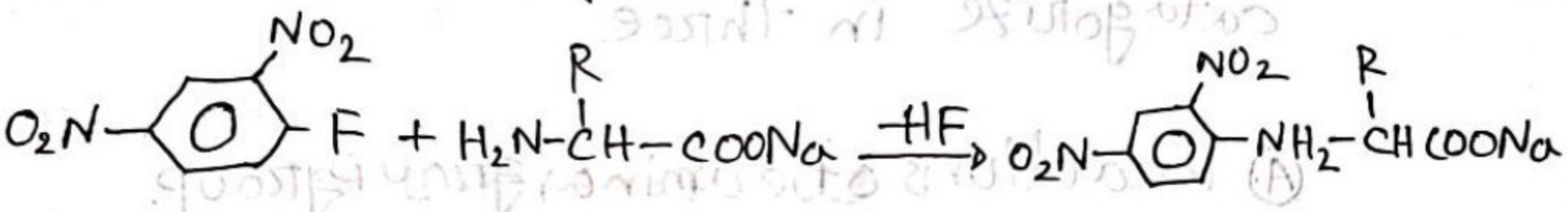
② Alkylation: Amino acids in alkaline solution react with alkyl halide to form N-alkyl substituted amino acid.



③ Acylation: Amino acids may be acetylated by means of acetyl chloride or acetic anhydride



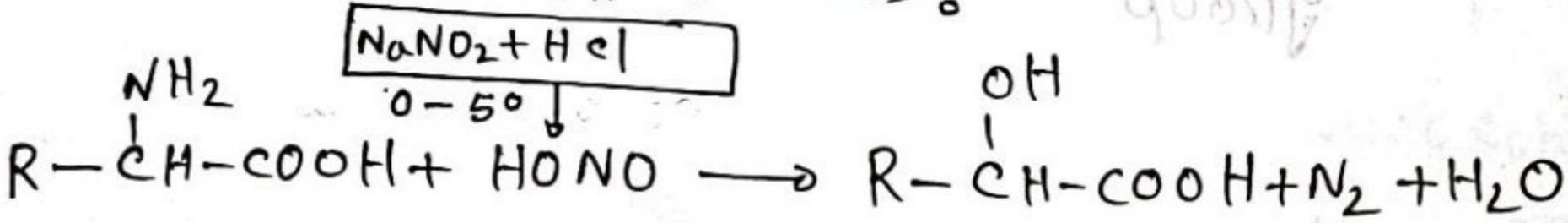
④ Reaction with sanger's reagent:



2,4 DNF or Sanger's reagent

2,4-Dinitrophenyl derivat.

⑤ Reaction with nitrous acids:

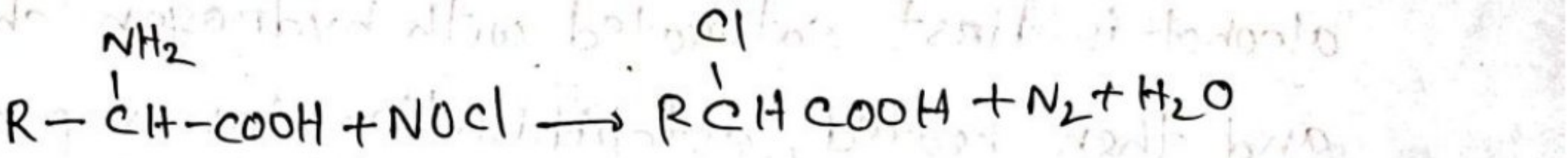


α-Hydroxy acid

This is the reaction for removing amino functional group.

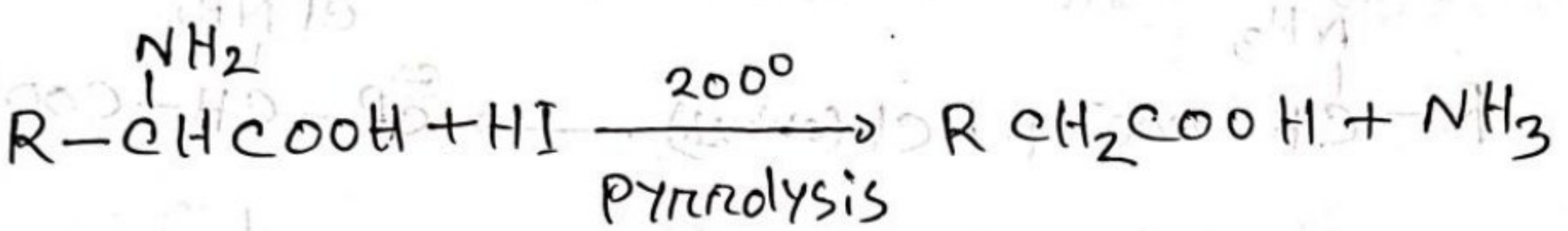
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⑥ Reaction with nitrosyl chloride or bromide:



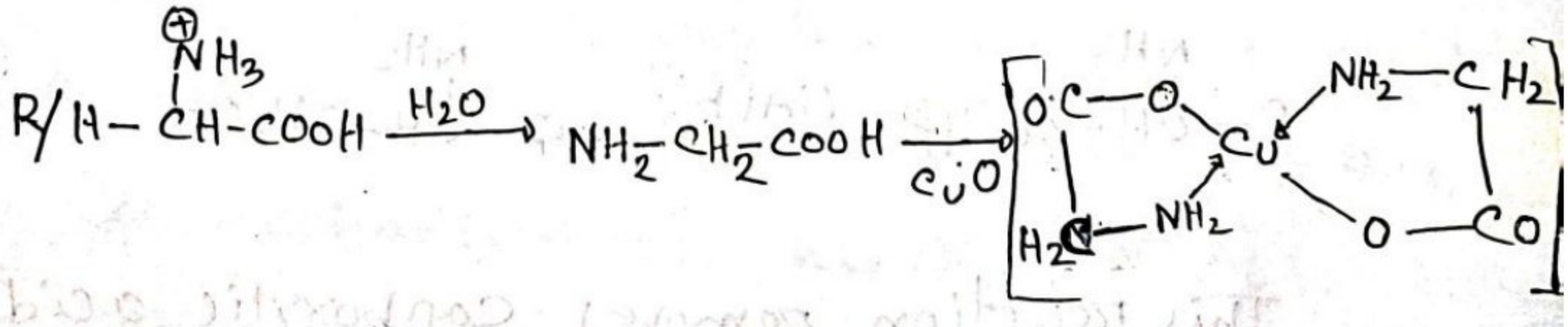
pH should be kept around 7.

⑦ Deamination: Amino acids on heating with hydrodic acid at 200°c gives the corresponding fatty acids with the elemination of ammonia.



Ⓑ Reactions due to the carboxyl group:

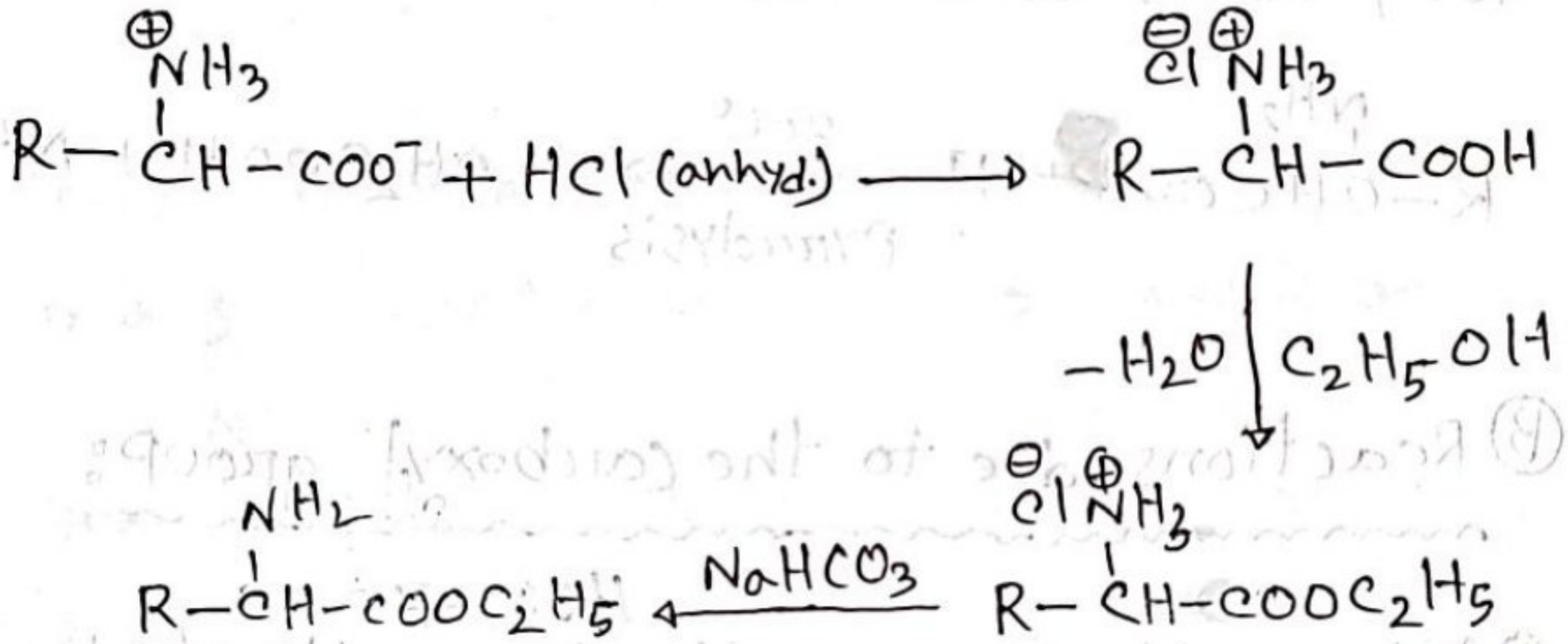
① Amino acids forms salts; the salts of heavy metals are chelate compounds, e.g, the copper salt of glycine (deep blue needles) is formed by heating copper oxide with an aqueous solution of glycine.



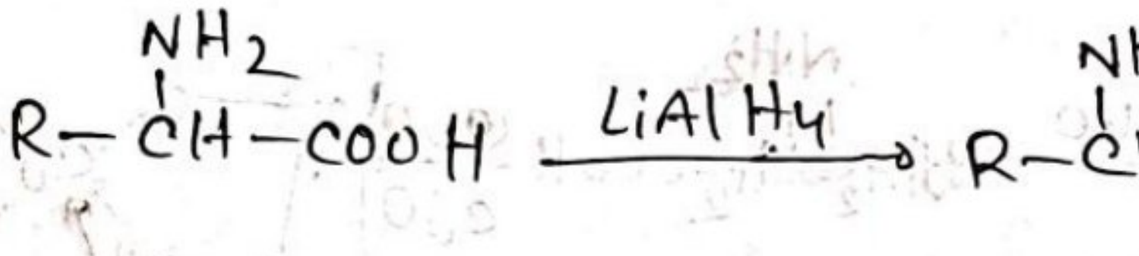
This is an identification reaction of amino acids.

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② Esterification: If the amino acid dissolved in alcohol is first saturated with hydrogen chloride and then heated, esterification takes place smoothly. This clearly indicates the alcohol does not react with zwitterion but with the acidic form to form ester salt. The free ester may then be obtained by the action of aqueous sodium carbonate on the ester salt.



③ Reduction of amino acids: Amino acids may be directly reduced to the amino alcohol with lithium aluminum hydride, and in this case no racemisation occurs.

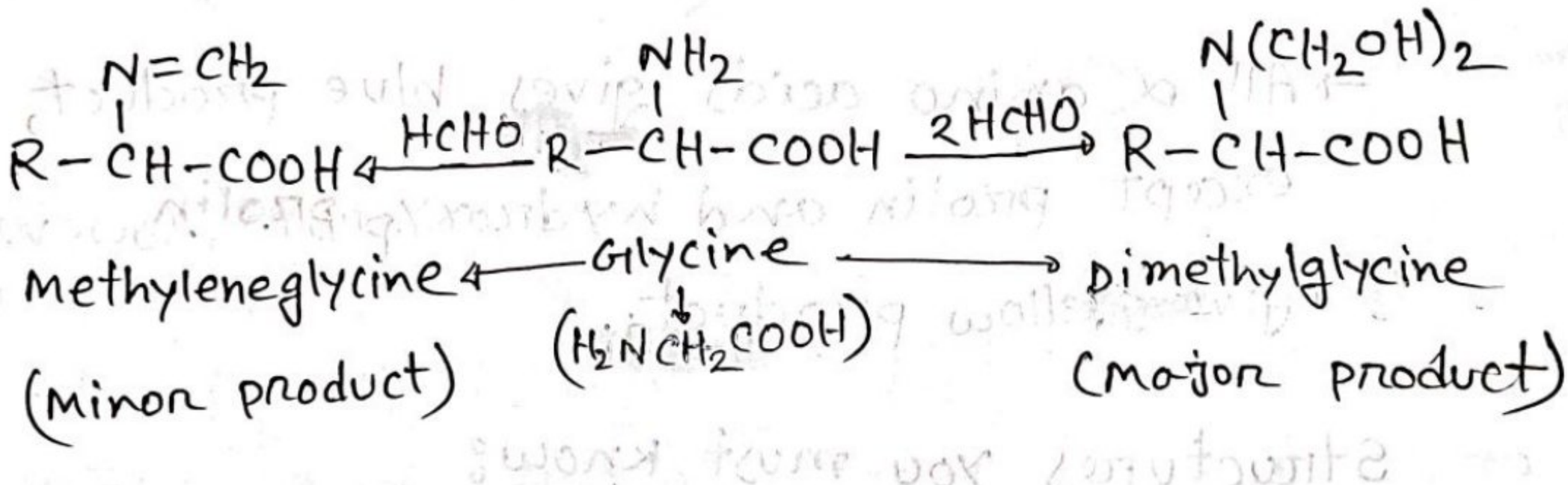


This reaction removes carboxylic acid group from amino acid.

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© Reactions due to both the amino and carboxyl groups:

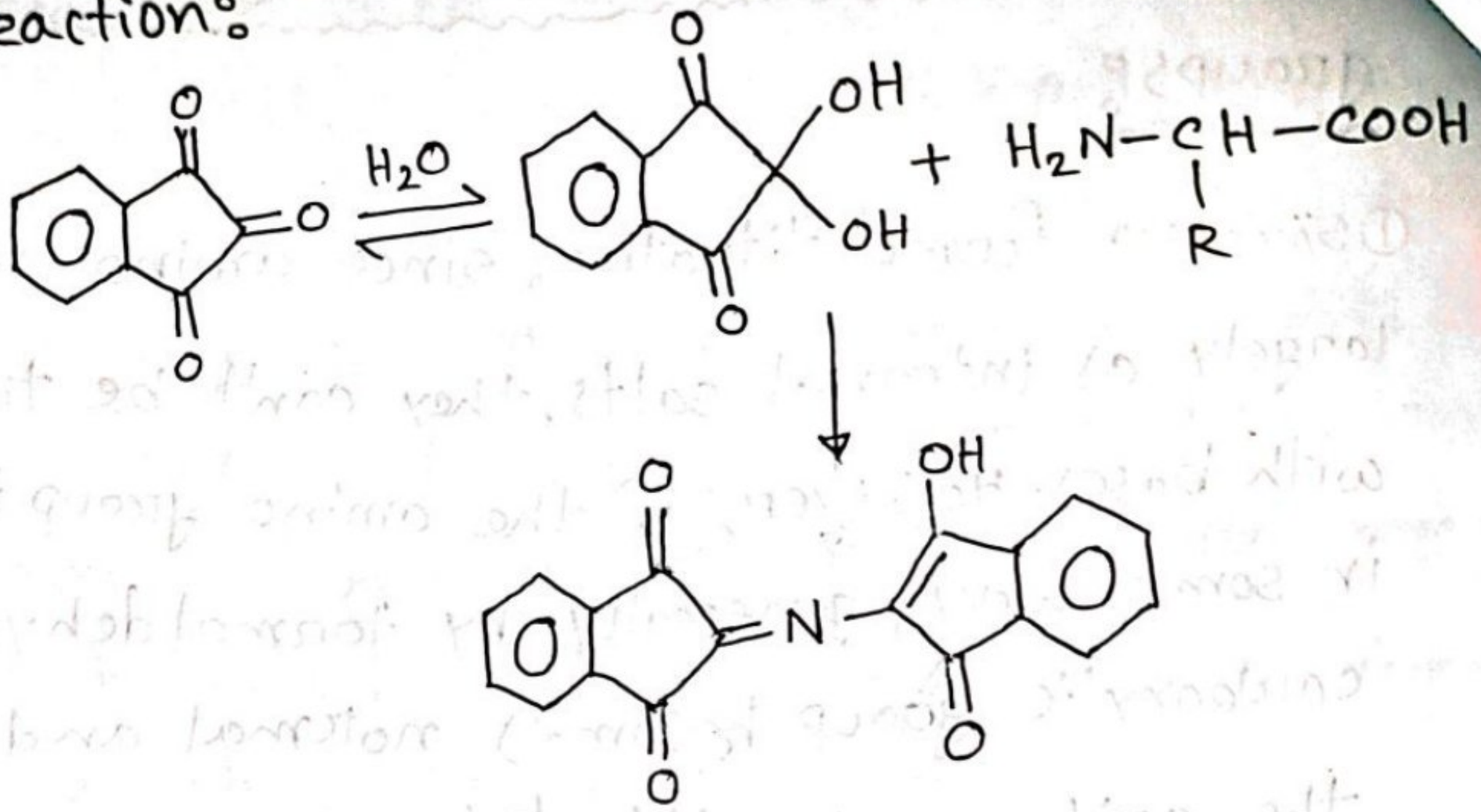
① Sörensen formol titration: since amino acid exist largely as internal salts, they can't be titrated with bases. However, if the amino group is blocked in some way, generally by formaldehyde; the carboxylic group becomes normal and thus the acid can be titrated using indicators.



② Ninhydrin reaction: Ninhydrin (indane-1,2,3-trione hydrate) reacts with amino acids to form a coloured (brilliant blue) product. This reaction is useful for the qualitative and quantitative estimation of amino acids. The mechanism of the reaction is uncertain.

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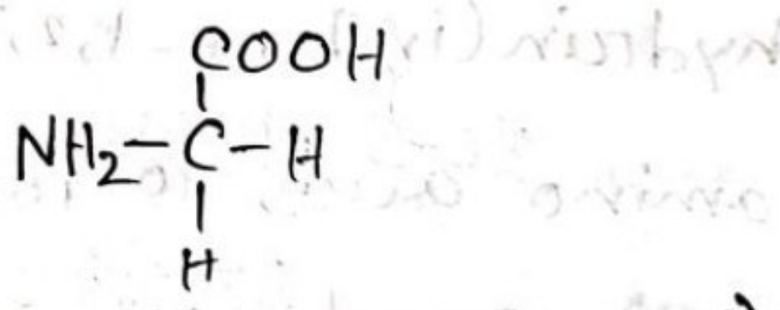
Reaction:



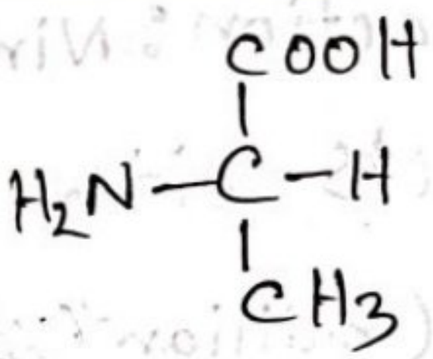
(Blue or violet)

→ All α amino acids gives blue product, except prolin and hydroxyprolin, however give yellow product.

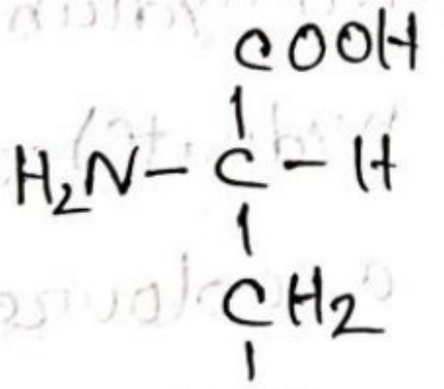
Structures you must know:



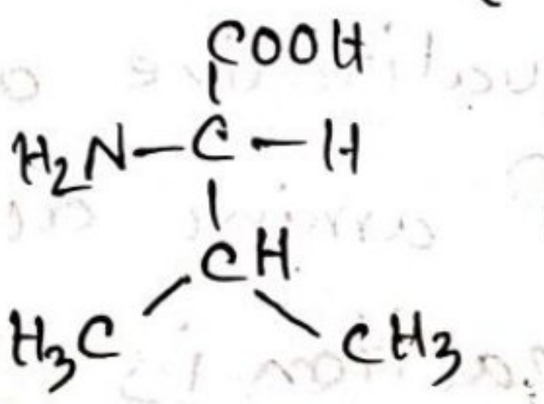
Glycine (α-aminoacetic acid)



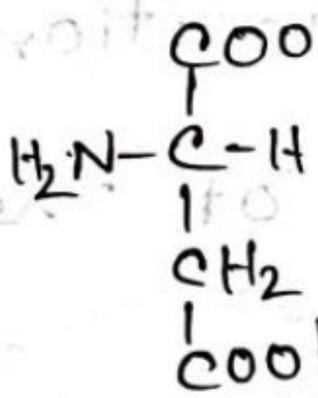
Alanine
(α-aminopropanoic acid)



Lysine



Valine
(α-aminoisovaleric acid)

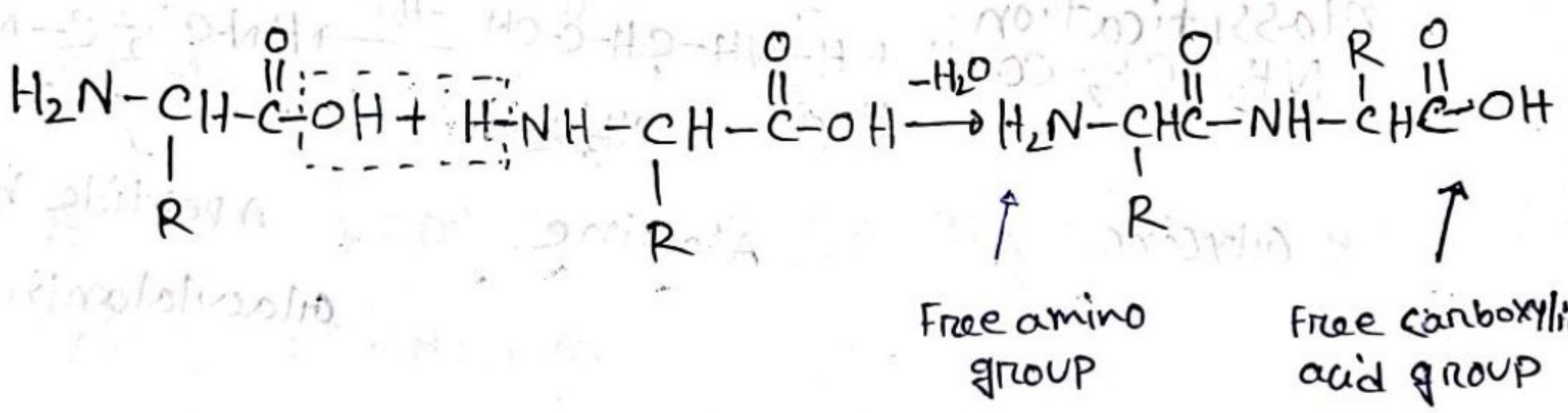


Aspartic acid
(α-aminosuccinic acid)

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Peptides and polypeptides

As the amino acid molecules contain both basic as well as acidic group, it might be expected that an intermolecular reaction may take place between the carboxyl group of one amino acid and the amino group of other amino acid, with the elemination of a molecule of water.



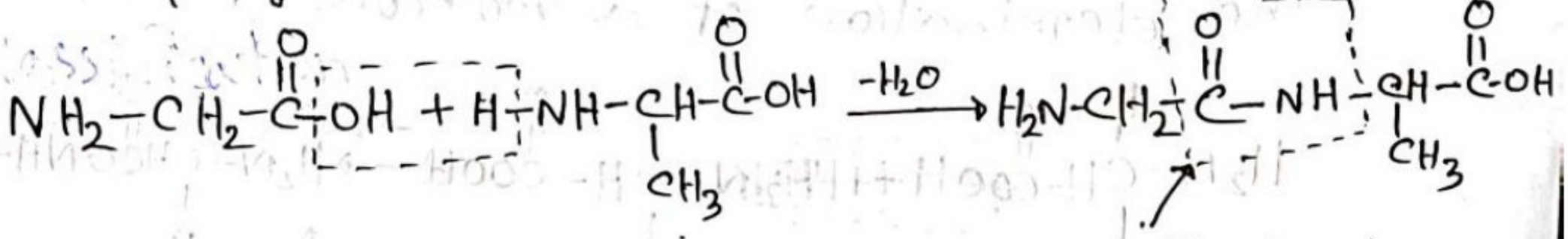
Since the resulting molecule still has a free amino and carboxylic acid group, it may react with other amino acids at either of the ends to give higher molecular weight product/condensation product.

→ The formulae of all resulting condensation products indi· that the every two amino acid are linked by means of –CO–NH– group, which is reffers as peptide bond.

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Definition of peptide: When two or more amino acid molecules are bonded together via peptide bond; -COOH from one amino acid, $-NH_2$ from anothe amino acid forming ($-\overset{O}{\overset{\|}{C}}-NH-$) peptide linkage, the compound is called peptide.

Example:



Glycine Alanine A peptide bond.

Glacylalanine (a dipeptide)

classification of peptides:

The peptides are classified in various group depending upon the number of amino acid residues. This can be - dipeptide, tripeptide, tetra-peptide or polypeptide.

→ peptide formation of two different amino acids can take place in two ways. For example,

Glycine + Alanine $\xrightarrow{-H_2O}$ Glacylanine.

Alanine + Glycine $\xrightarrow{-H_2O}$ Alanylglycine.

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Depending on the product when a peptide is hydrolysed, the peptide can be classified in two types.

① Homomeric;

on hydrolysis it gives only amino acids.

② Heteromeric; Hydrolysis gives amino acids as well as other products.

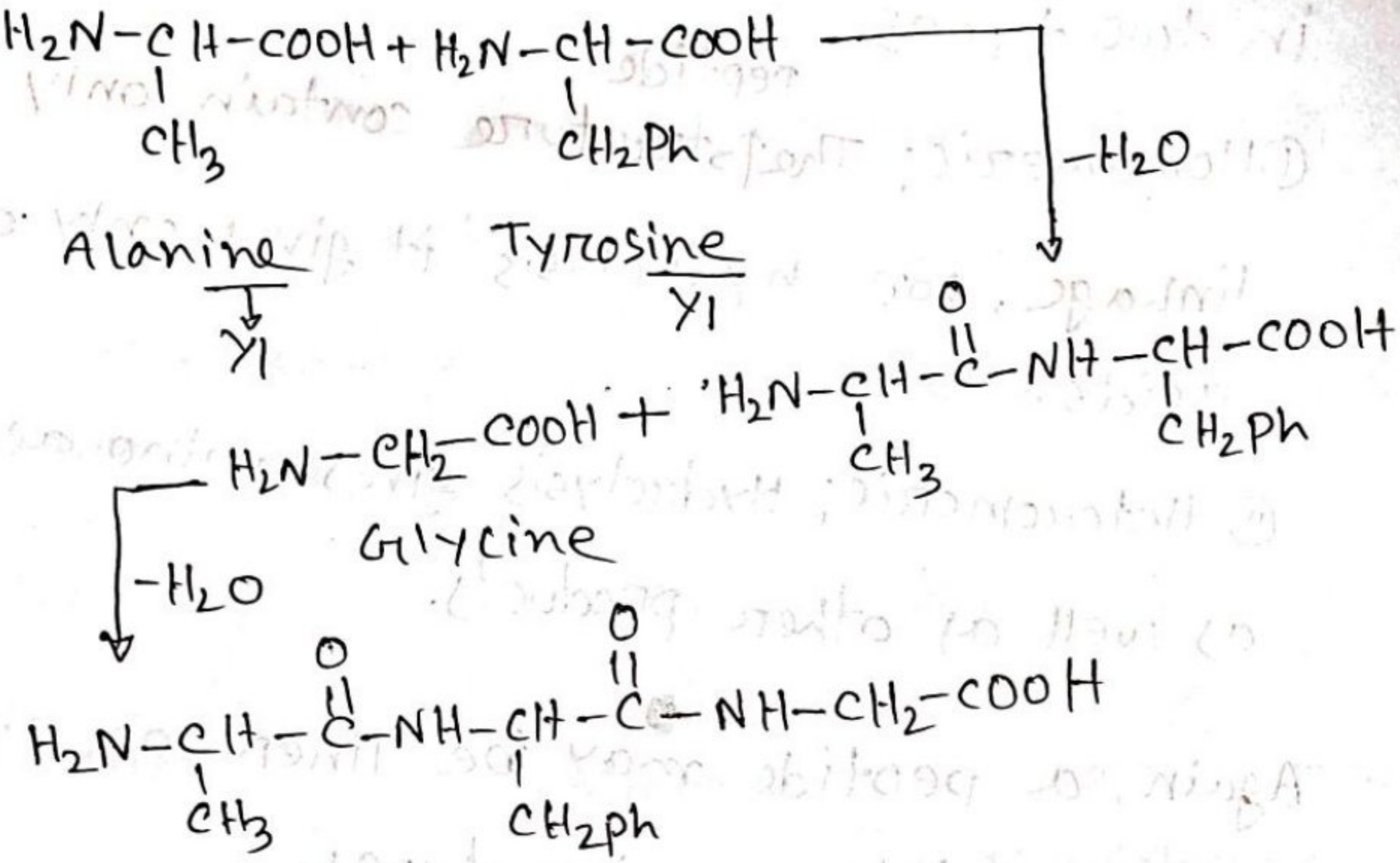
Again, a peptide may be liner or cyclic. cyclic peptide are two types.

① Homodetic: structure contains only peptide linkage.

② Heterodetic: structure contain both peptide linkage and other linkage.

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Nomenclature of peptides:



Alanyltyrosylglycine

Full name, cause it has free -COOH. According to joining order other amino acids short forms are used in the name.

→ Replace (ine) from the end of every name with (Yl).

→ If we know the sequence then

Common abbreviation

① Alanyl tyrosyl glycyl
② Ala-tyr-gly
③ Ala.tyr.gly
④ Ala→tyr→gly

only use (→) when we know the sequence.

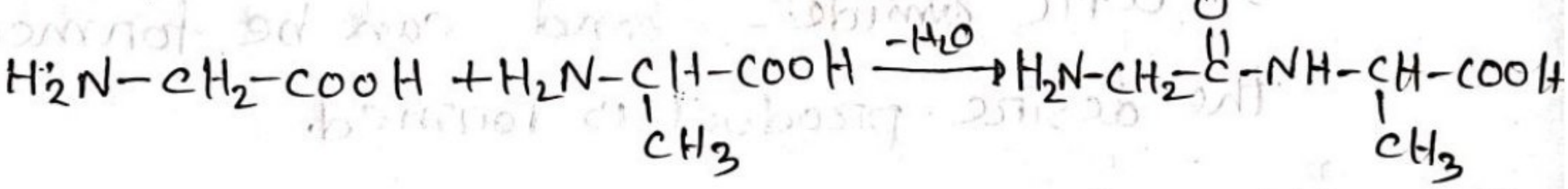
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Synthesis of peptides:

* why direct method is not convinient for peptide synthesis?

Ans: In direct method, mixture of product is obtained, cyclic and chain compounds of different pattern may form. Availabity of desire product is rare.

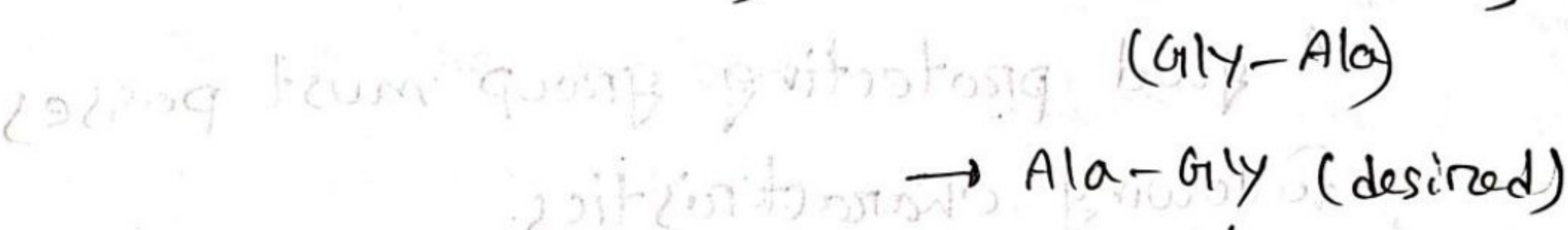
Lets assume Alanylglycine is the desire product. In direct method a wide variety of products may form-



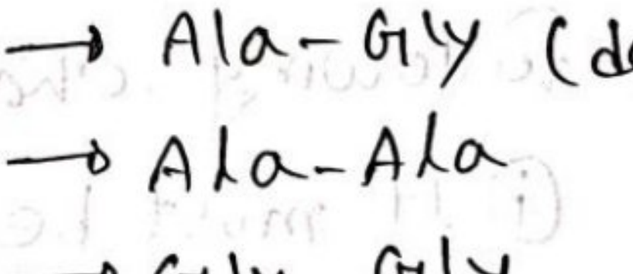
(Gly-Ala)

→ Ala-Gly (desired)
→ Ala-Ala
→ Gly-Gly
→ Ala-Gly-Ala-----
→ many more

Ring compound also may form,



Due to low yeild and difficulities in separation the direct method is not convinient.



cyclic amide (Lactum)

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General principles of polypeptide synthesis:

1) Protection of amino group: This is mandatory to prevent polymerization of amino acids and to minimize undesirable side reactions during the synthetic process. Proper protecting group manipulation stratagies can maximize the yield of the desired product. By protecting the amine group of one component and carboxylic acid group of the other, a specific peptide bond can be formed, thus the desire product is formed.

A good protecting group must posses the following characteristics

(i) It must be easily attachable to the amino group without causing racemization of an optically active acid.

(ii) It must be inert during reactions when peptide bonds are formed.

(iii) It must be easily removeable without effecting the peptide linkage or sensitive functional groups on amino acid side chain.

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2) Activation of carboxyl group:- The carboxylic group of the amino group protected amino acid is activated by converting it into acid chloride, mixed anhydride, acid azide or P-nirophenyl ester. ($SOCl_2$, PCl_5, PCl_3 can be used)

3) The coupling step:- This is the actual step where peptide bond formation takes place. The formation of a peptide bond is generally achived by the nucleophilic attack of free amino group of an carboxyl group protected ester on the activated carboxyl acid carbon of the N-protected amino acid.

→ The more electronegative the activing group(leaving group) the faster will be the peptide bond formation.

→ In general, the coupling step consists in treating equimolar mixture of the amino and conboxyl component in a solvent of low polarity at $-10°$ (to avoid the loss of optical activity) for few hours or days.

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(iv) The sidechain of trifunctional amino acids (e.g. Cys, Tyr, Asp, Glu etc) must be protected by special blocking groups, which are stable to the condition at which protecting group of amino or carboxyl group is cleaved.

The important protecting groups which fullfill all above conditions –

→ Amino protecting groups:

Benzyloxycarbonyl ($C_6H_5-CH_2-O-\overset{O}{\overset{\|}{C}}-$)

tert-Butyloxycarbonyl ($(CH_3)_3C-O-\overset{O}{\overset{\|}{C}}-$)

Triphenyl methyl ($(Ph)_3C-$)

Phthalyl ($-CO-C_6H_4-CO-$)

Tosyl ($R_2N-\overset{O}{\overset{\|}{S}}(=O)-C_6H_5$)

→ Carboxyl protecting groups: The usual method of protecting a carboxyl group is esterification, the common esters being methyl, ethyl, benzyl & t-butyl.

$$R-\overset{O}{\overset{\|}{C}}-OH + HO-CH_2-C_6H_5 \xrightleftharpoons{H^+} R-\overset{O}{\overset{\|}{C}}-O-CH_2-C_6H_5$$

(protected; deprotecting: $NaOH, H_2O$)

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4) Removal of the protecting group:

After forming polypeptide chain of desire length, the protecting group is removed to give final product. Benzyloxycarbonyl ($C_6H_5-CH_2-O-\overset{O}{\overset{\|}{C}}-$) group is removed by catalytic reduction. (H_2-Pd)

→ If the amino-acid contains sulphur (cystine), then catalytic reduction can't be used since sulphur poisons the catalyst.

→ Then HBr in (acetic acid or nitromethane) is convinient.

Synthesis of Glycylalanine

<u>Step 1:</u> Protection of N-terminal amino group.

$$C_6H_5-CH_2-OH + COCl_2 \longrightarrow C_6H_5CH_2OCOCl$$

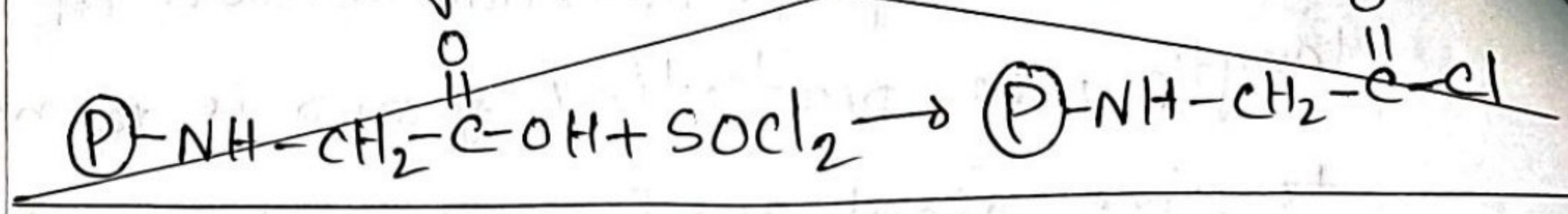
(Benzyloxycarbonyl chloride)
(protecting agent)

$$C_6H_5CH_2OCOCl + H_2N-CH_2-\overset{O}{\overset{\|}{C}}-OH \longrightarrow C_6H_5CH_2O\overset{O}{\overset{\|}{C}}-NHCH_2COOH$$

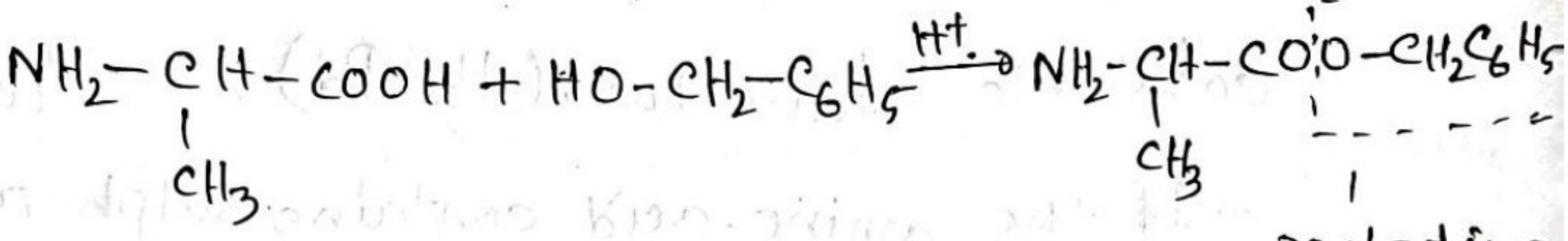
Glycine

$C_6H_5CH_2O\overset{O}{\overset{\|}{C}}-$: protecting group (P)

~~Step 2°~~ Activati carboxyl group of amino group protected amino acid.

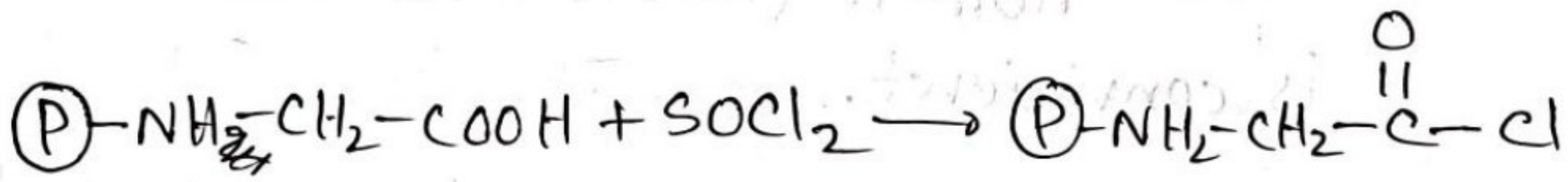


Protection of c terminal carboxyl group:



Alanine (An ester) protecting group

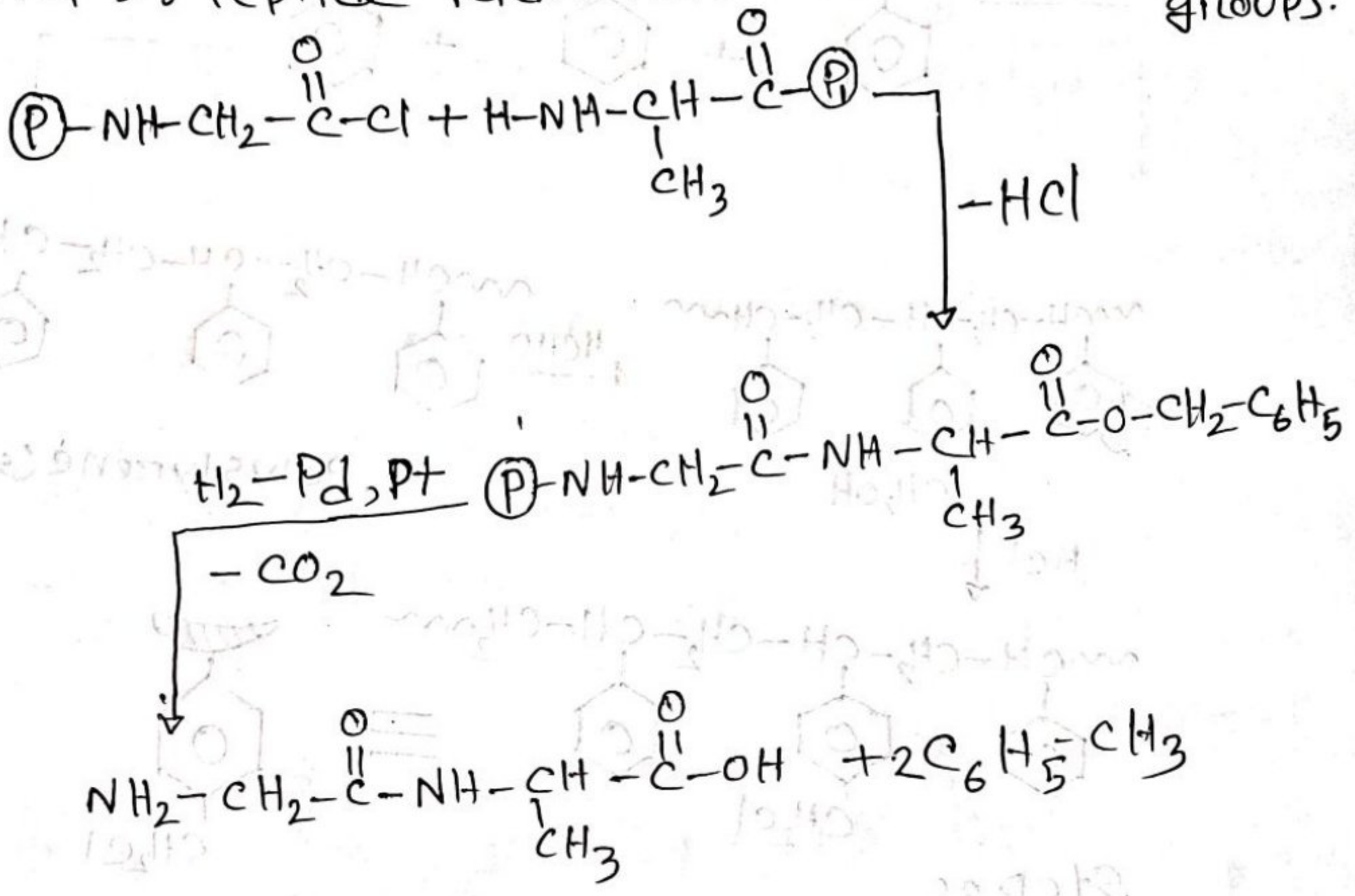
Step 2°: Activation of the -COOH group of the amino group protected amino acid



→ Here Cl is a very good leaving group, during a peptide bond formation a molecule of HCl is released. Since HCl is gaseus it leaves the reaction medium, so we get better yeild (than using other activating agent which may forms H_2O when leaves).
we could have used PCl_5 or PCl_3.

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Step 3: Peptide formation and removal of protecting groups.

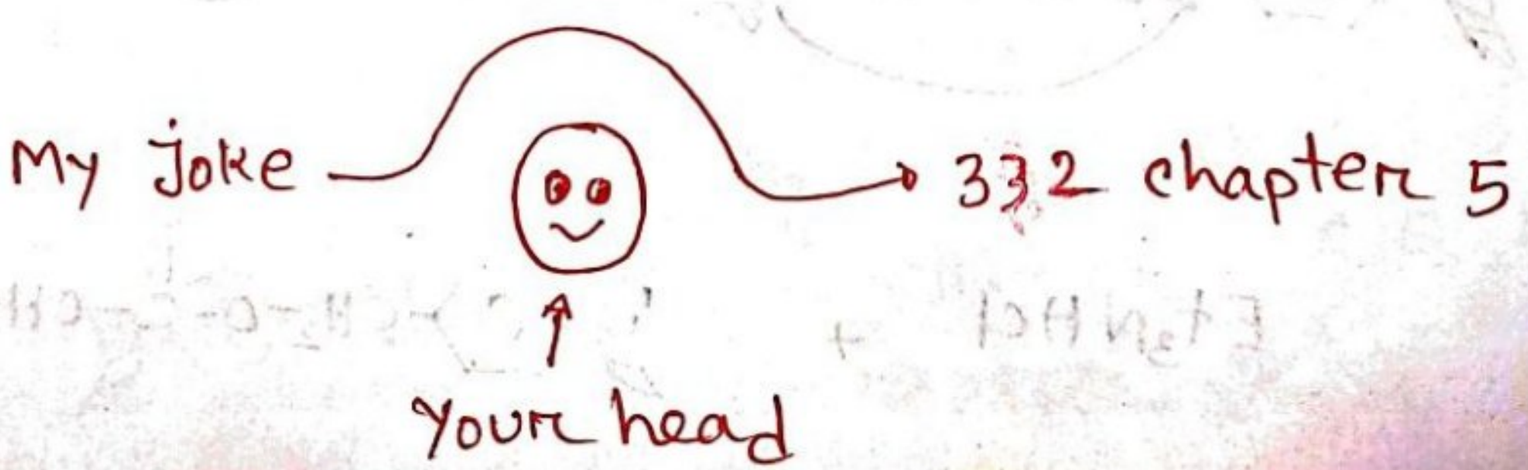


Glycylalanine

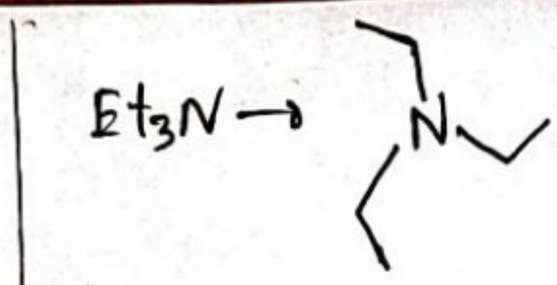
Merryfield solid phase peptide synthesis

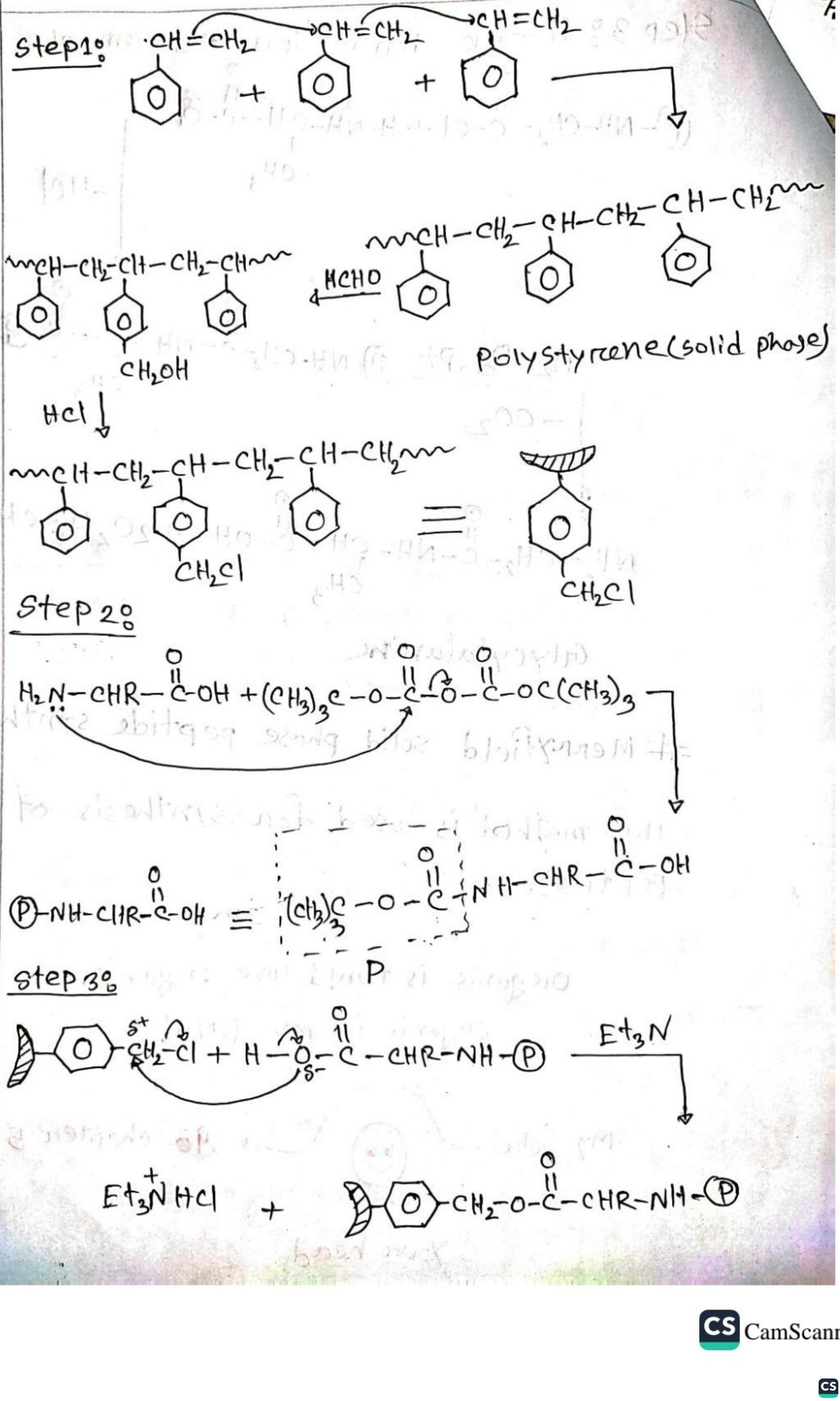
This method is used for synthesis of larger peptides.

Organic is fun! I love organic.
Organic in my DNA!



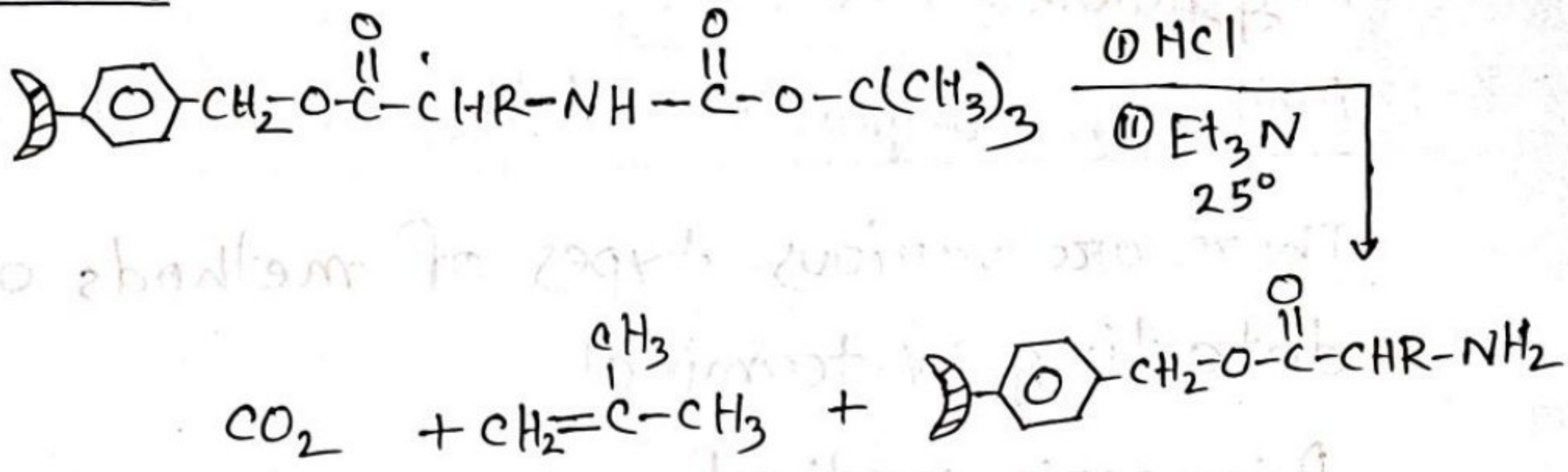
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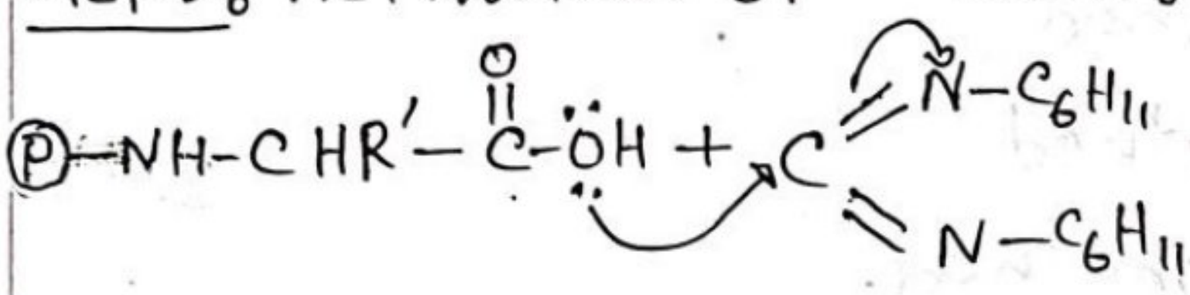

Step1°
CH=CH2
HCHO
CH2OH
HCl
CH2Cl
polystyrene(solid phase)
Step2°
Step3°
Et3N
Et3NHCl

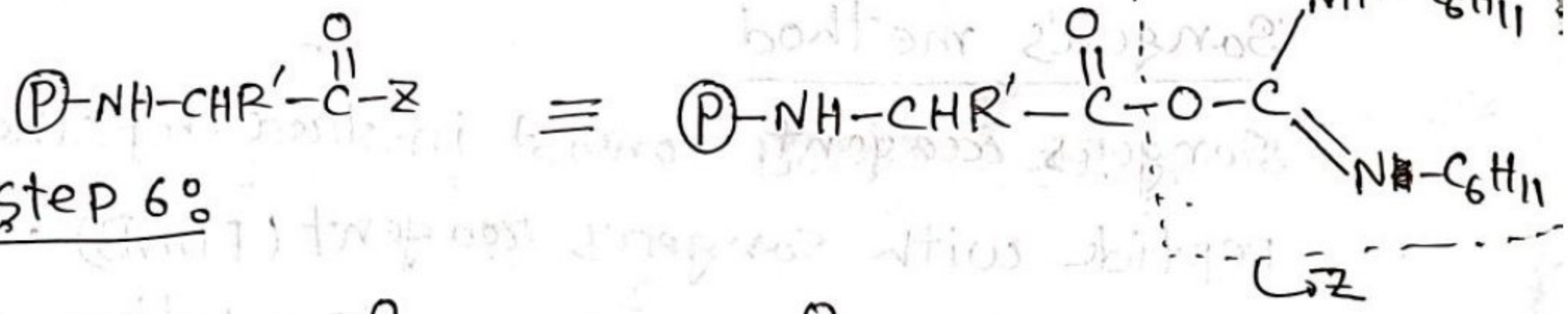
Note: To prepare/synthesis Ala-Phe, we just input the valve of R=CH_3, R'=$C_6H_5CH_2$

Step 4°

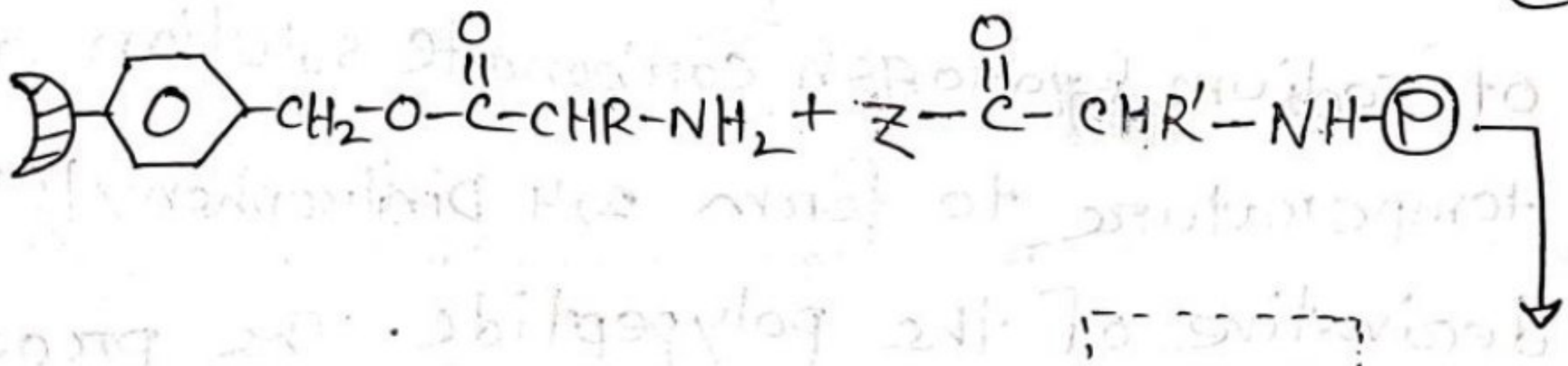


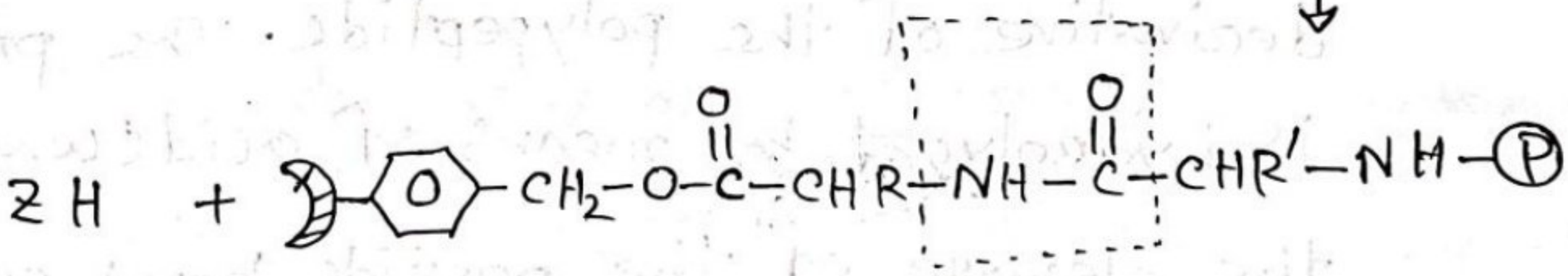
Step 5° Activation of $-COOH$ °



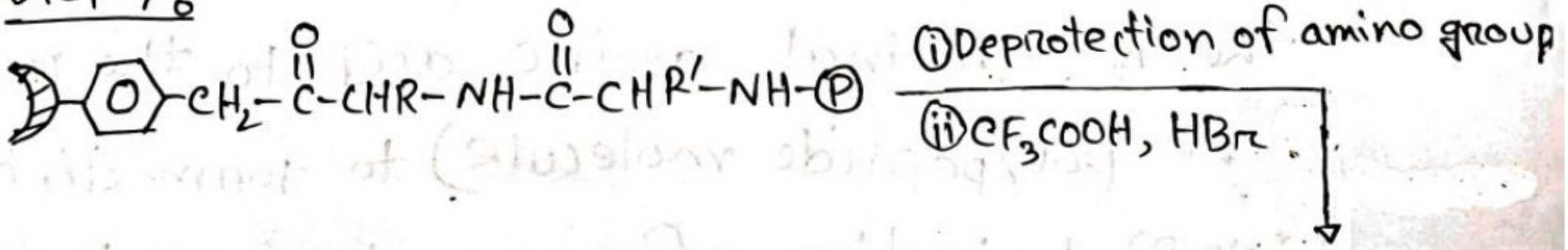


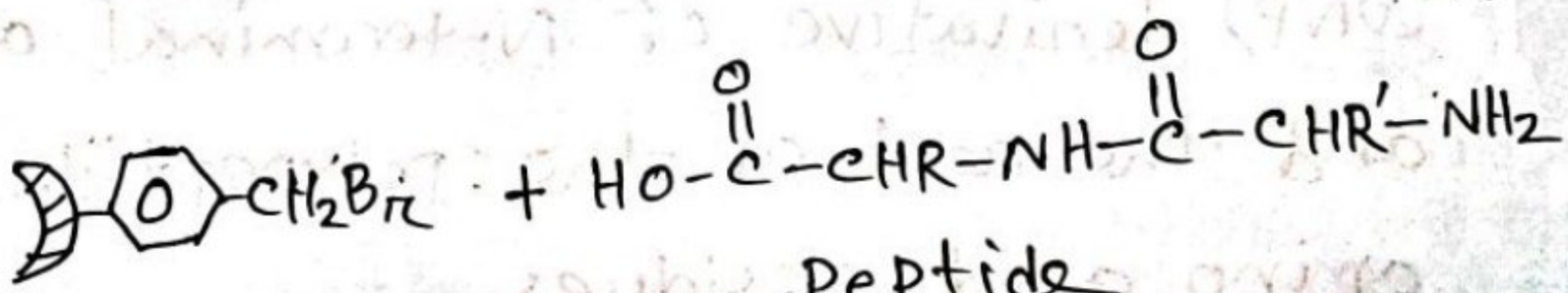
Step 6°





Step 7°





Peptide

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End group analysis

Amino-End Degradation / Dtection of N-terminal

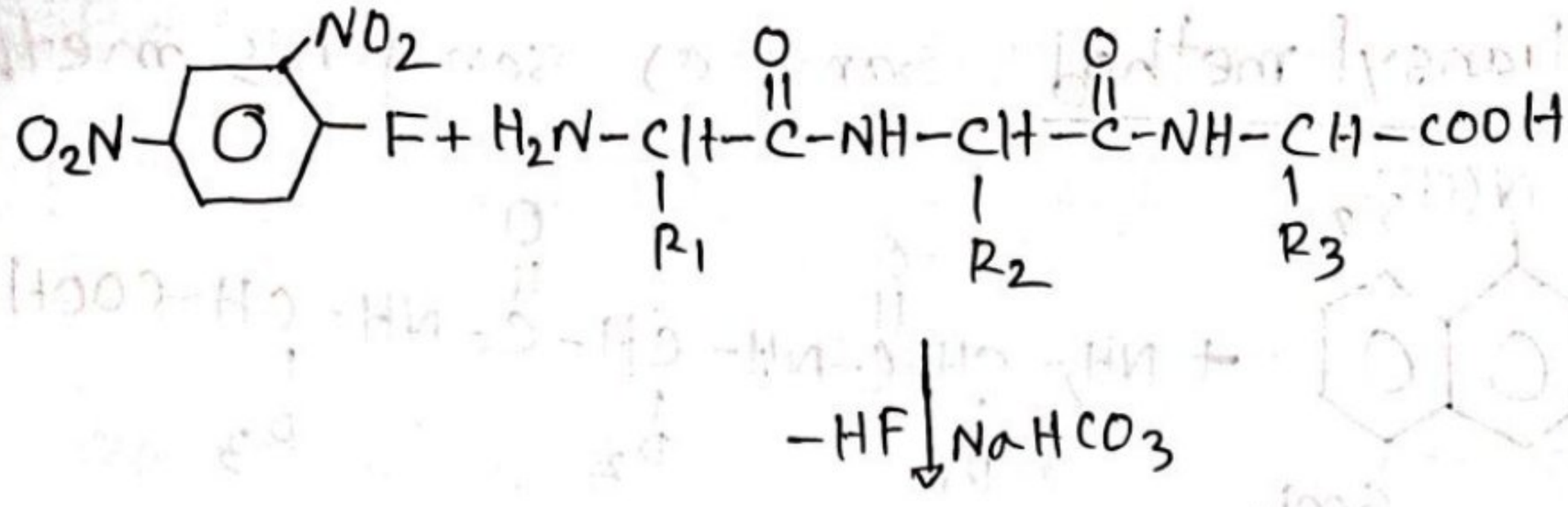
There are various types of methods of detecting N-terminal

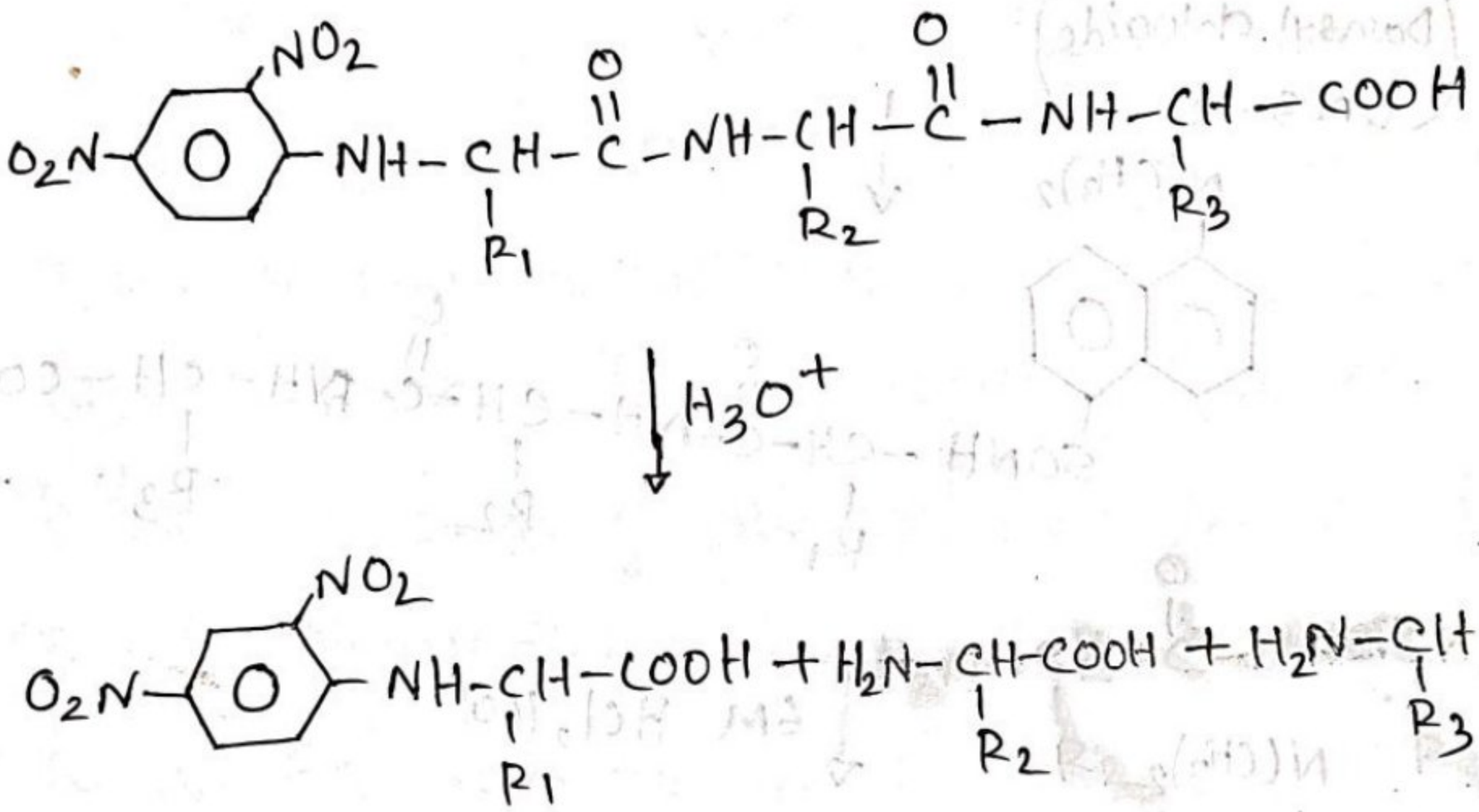
1) Sanger's method
2) Dansyl method
3) Edman method
4) Enzymatic method.

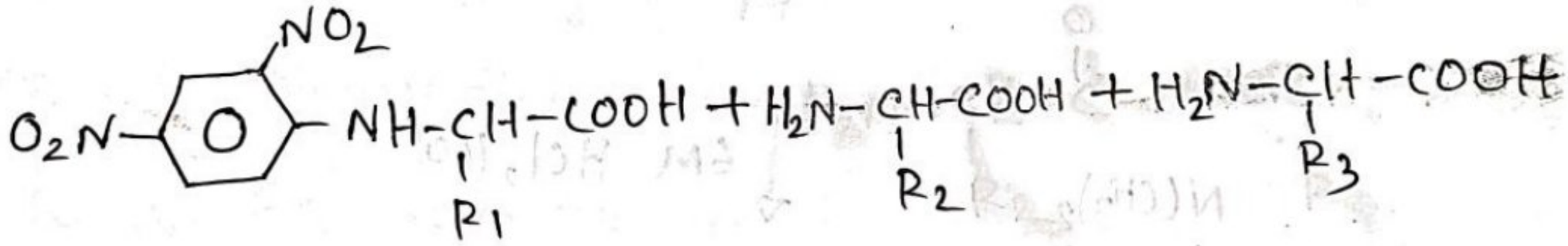
Sanger's method

Sanger's method consist in treating the polypeptide with sanger's reagent (FDNB) in presence of sodium hydrogen carbonate solution at room temperature to form 2,4 Dinitrophenyl (DNP) derivative of the polypeptide. The product is hydrolysed by means of acid (which causes the clevage of the peptide bond connecting the N-terminal amino acid to the rest of the polypeptide molecule) to form dinitrophenyl (DNP) derivative of N-terminal amino acid and the rest of the polypeptide molecule or amino acid residues.

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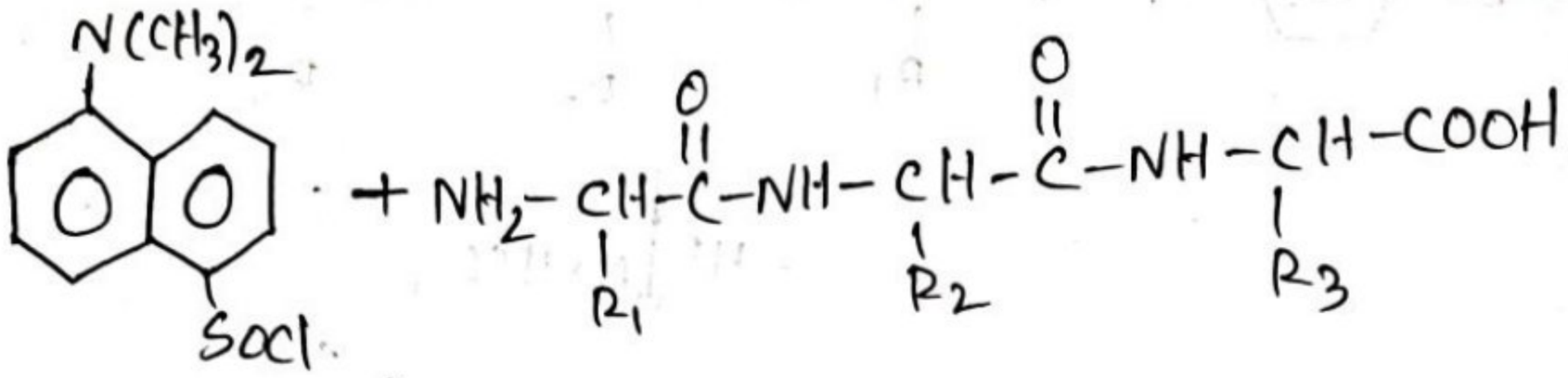


DNP derivative identified via TLC

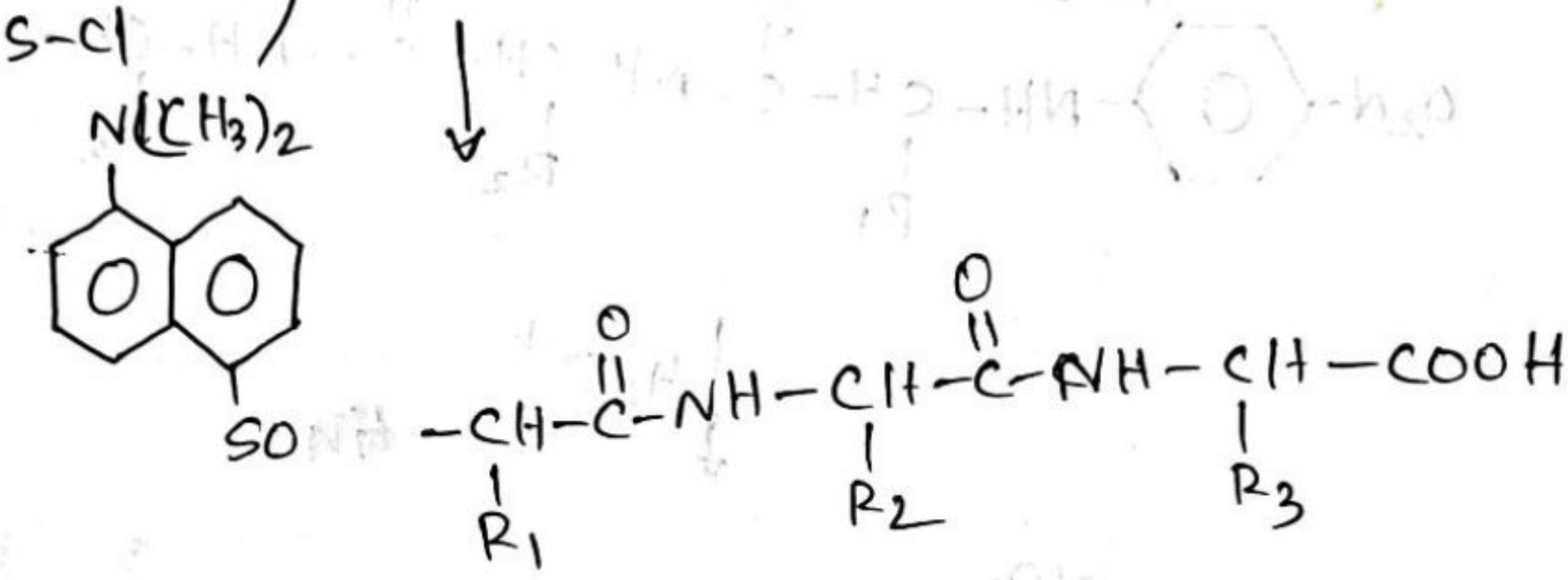
After isolating the DNP derivative, it is identified by chromatography or spectroscopy. Thus N-terminal amino acid in a polypeptide or protein can be identified.

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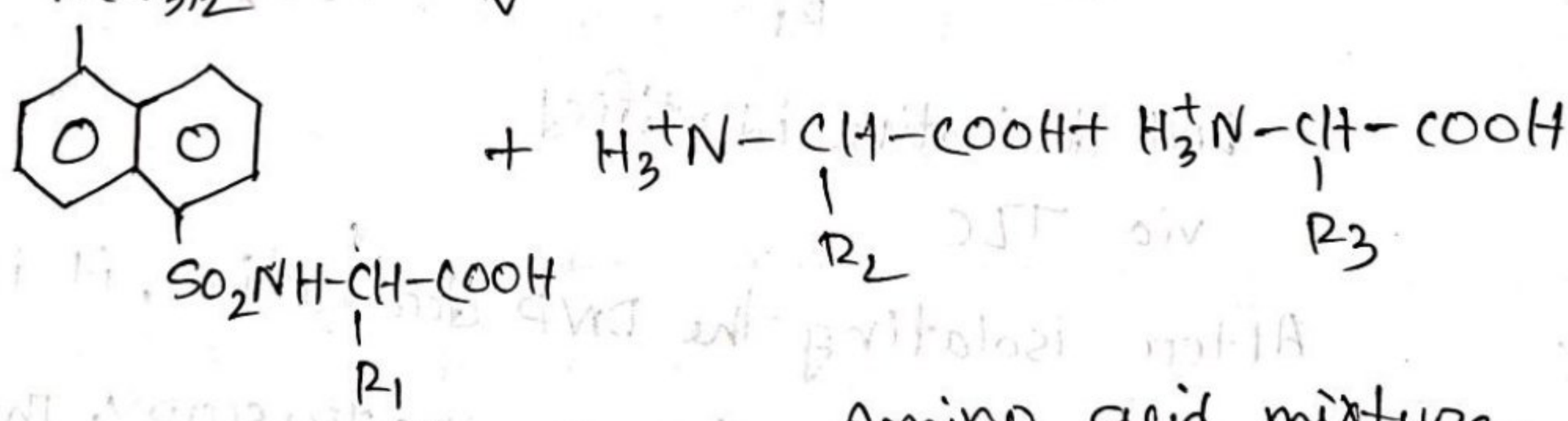
2) Dansyl method : Same as sanger's method.



(Dansyl chloride
DNS-Cl)



6M HCl, 100°



Dansyl derivative of amino acid identified via TLC.

Amino acid mixture-

This one is not in the class note

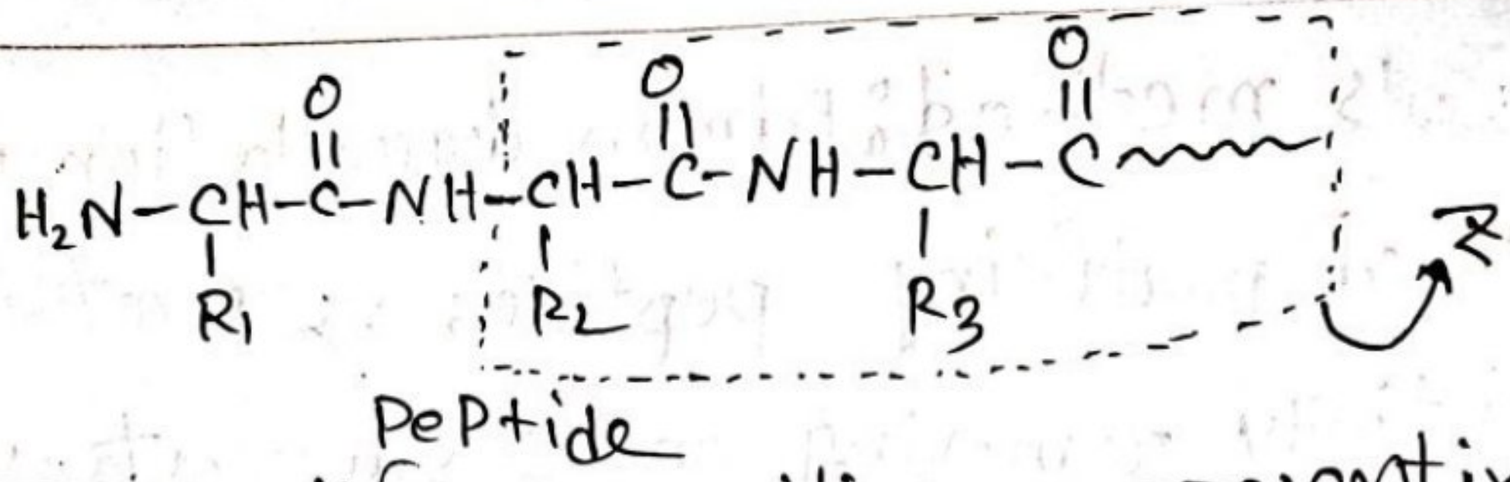
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3) Edman's method: Edman degradation is the process of purifying peptide or protein sequentially removing one residue at a time from the amino end of a peptide. In two methods discussed above the overall sequencing of a peptide is damaged by hydrolyzing conditions. But Edmans method dosnot damage the overall sequence. This is doen by adding phenyl isocyanate, which create) phenylthiocarbamoyl derivative with the N-terminal. The N-terminal is then cleaved under less harsh acidic condition creating a cyclic compound of phenylthio-hydantoin (PTH-amino acid). This does not damage the overall peptide sequence. This method can be repeated for the rest of the residues, separating one residue at a time.

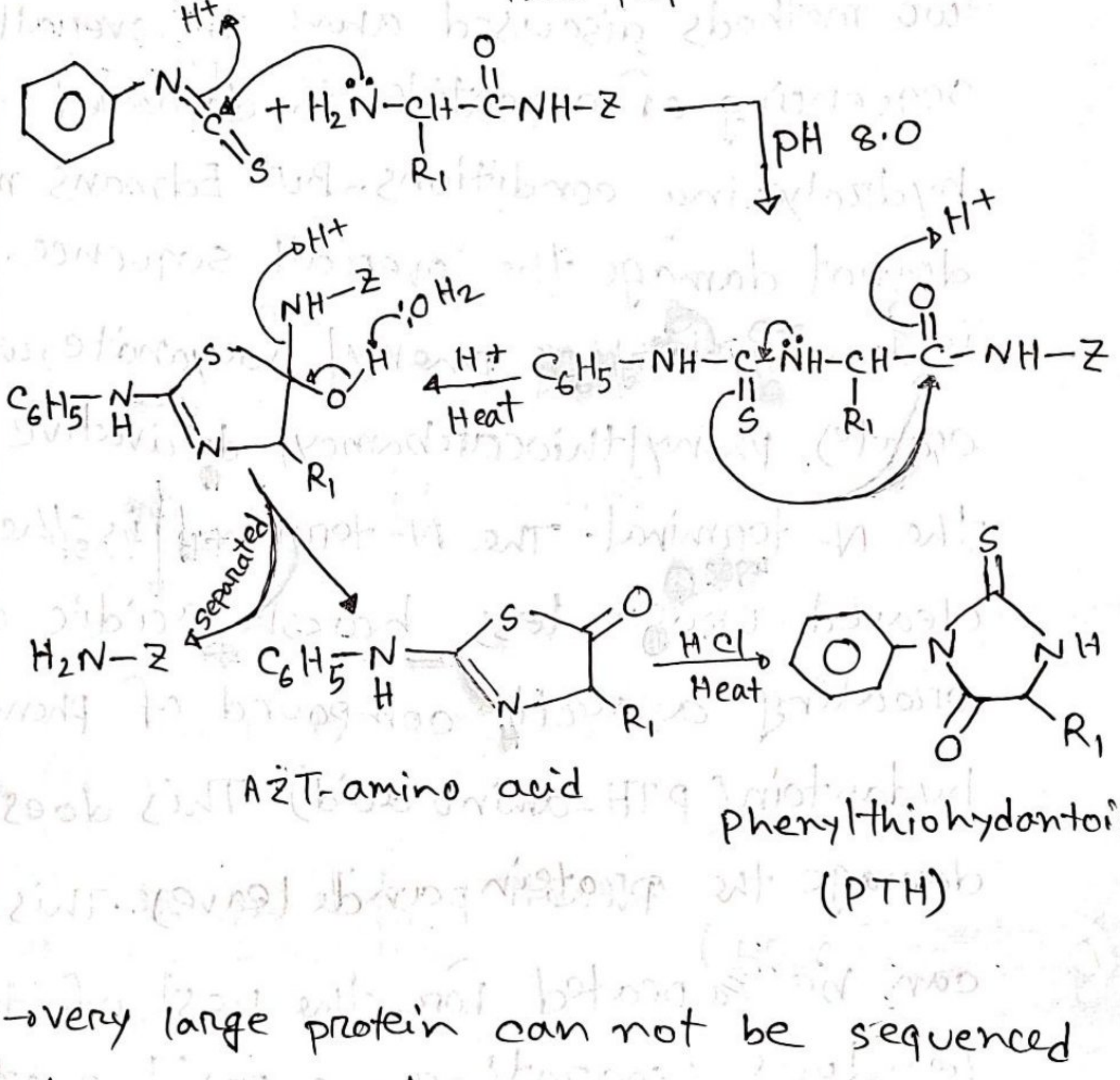
This one is very important for final

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(without mechanism)



To simplify the writing, representing the boxed portion of the peptide chain as Z.

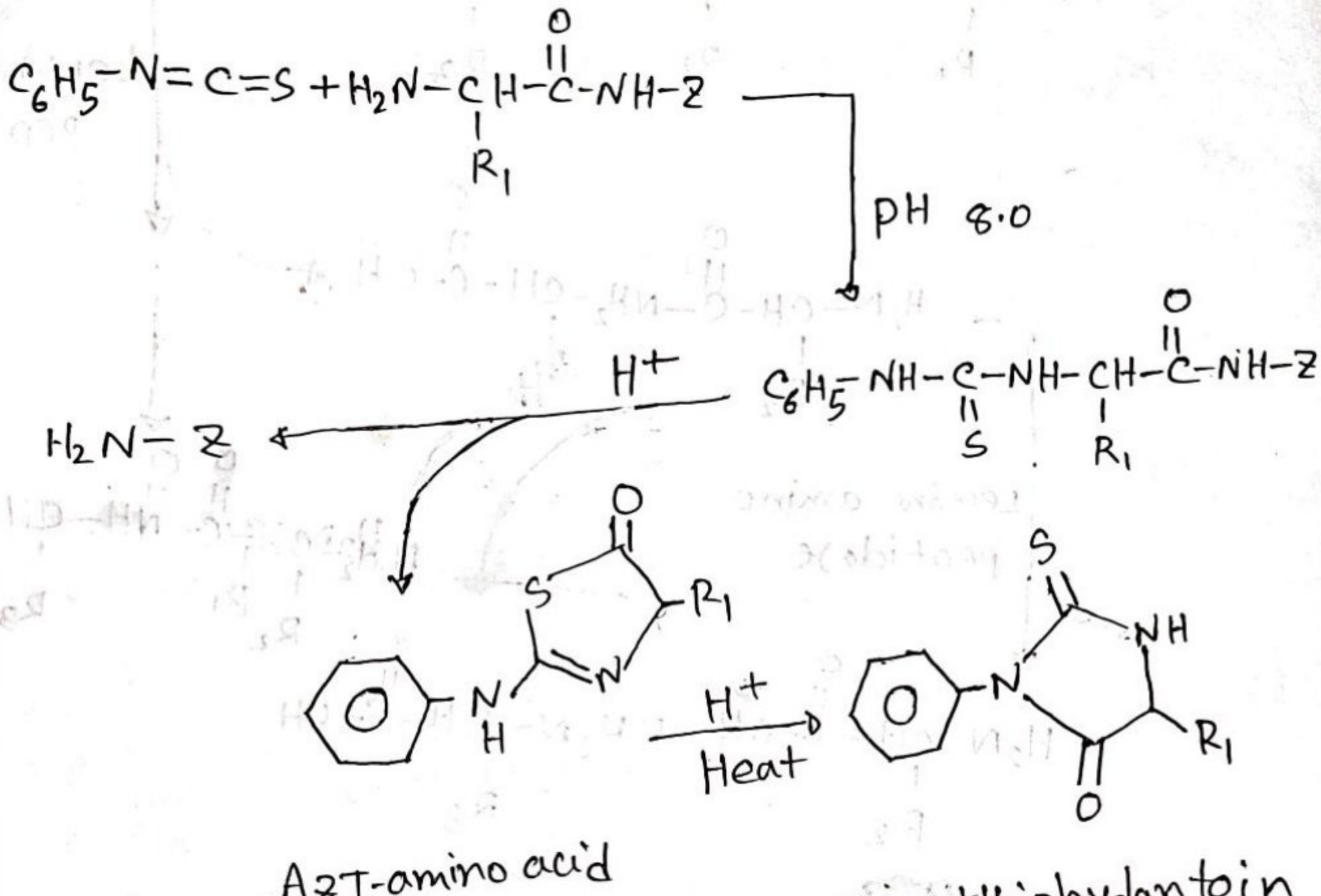


See next page for easy one!

→very large protein can not be sequenced by Edmans method because of less perfect efficiency. So larger proteins are cleaved into smaller first using enzymatic method.

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In more simple way!

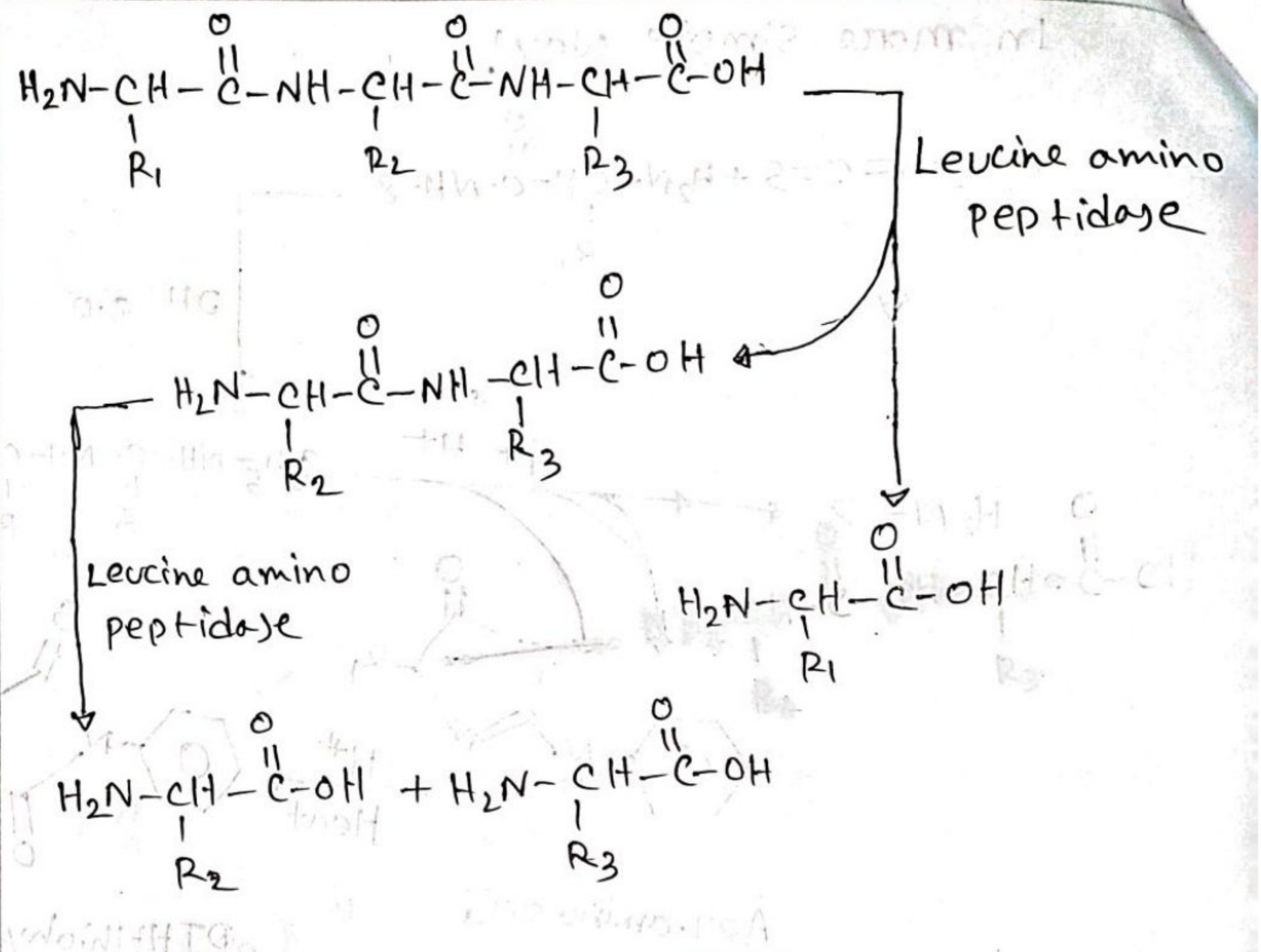


PTH maybe separated and identified using paper chromatography.

4) Enzymetic method:

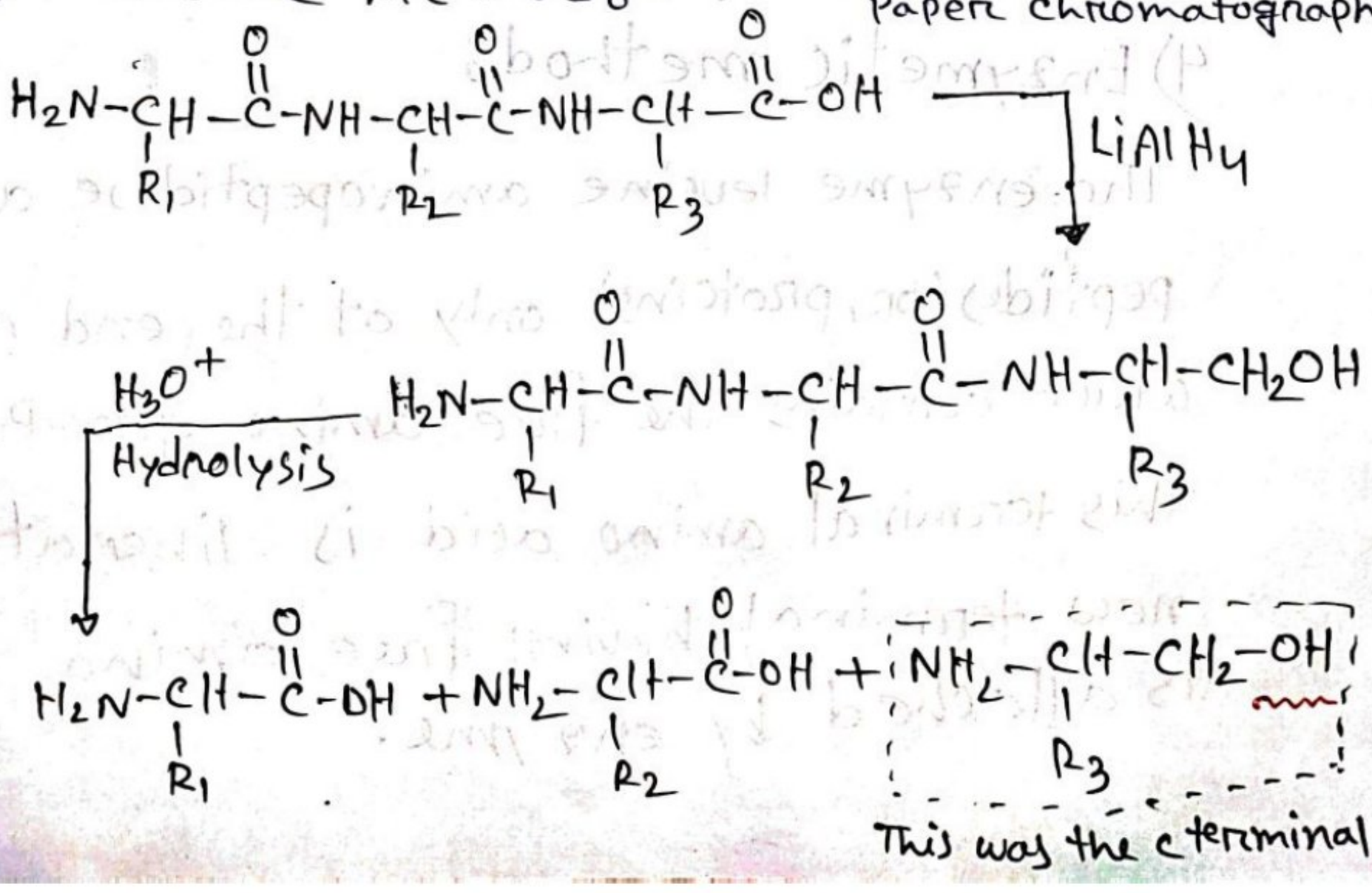
The enzyme leucine aminopeptidase attacks peptides (or, proteins) only at the end at which contains the free amino group. when this terminal amino acid is liberated, the new terminal having free amino group is attacked by enzyme.

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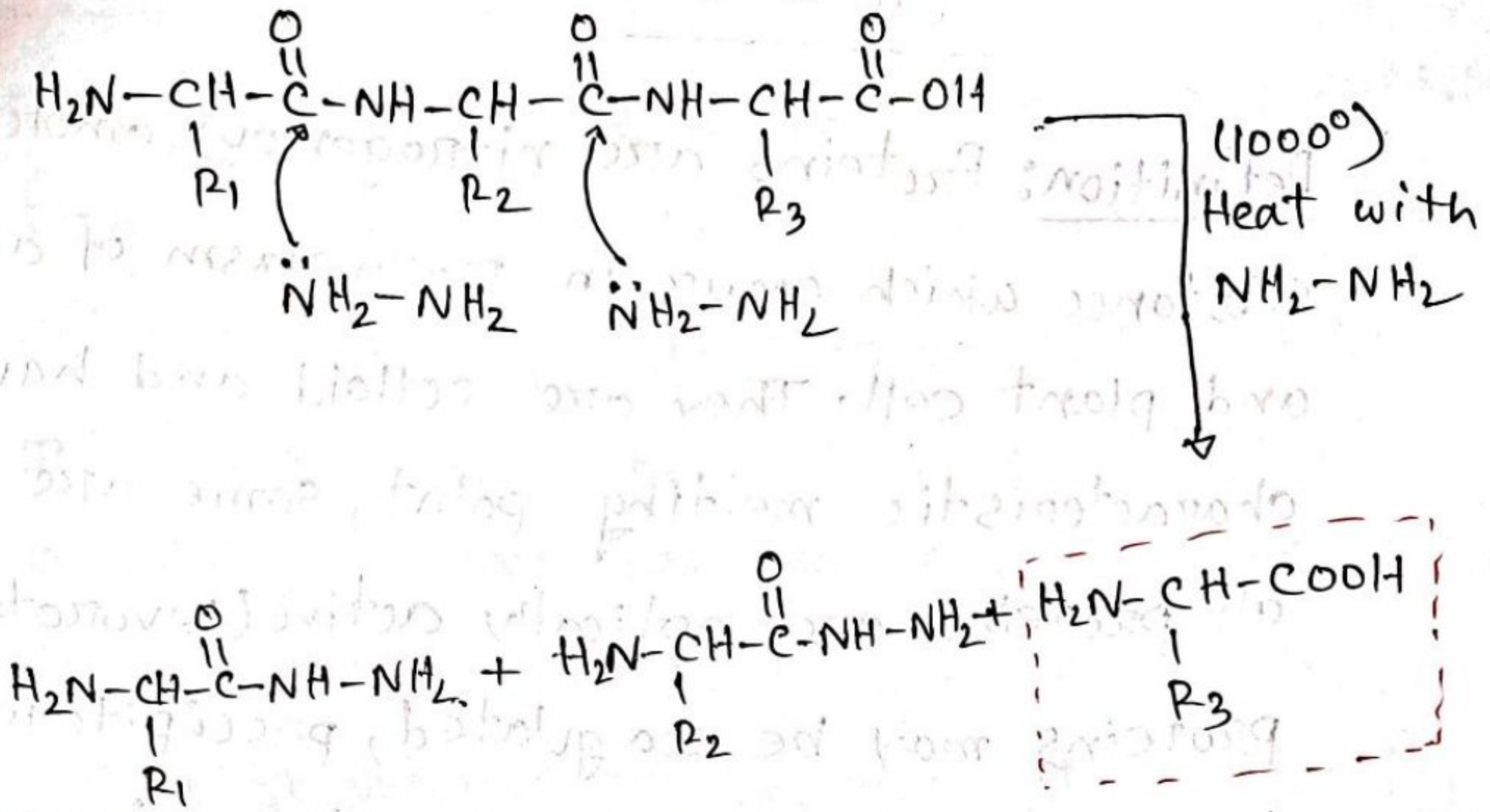


Determining C-terminal amino acid

1) Reductive method (separated and identified by Paper chromatography)



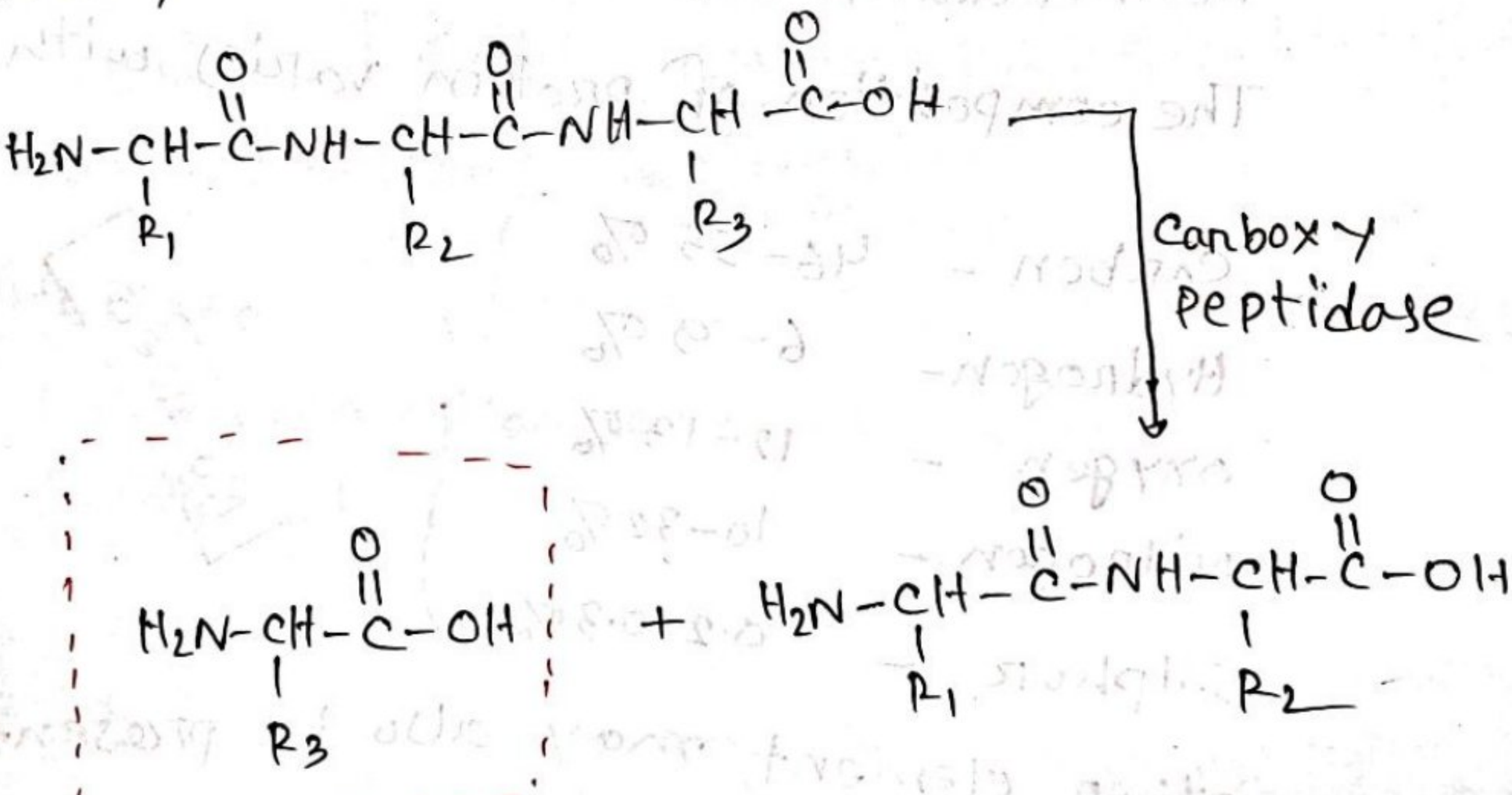
2) Hydrazinolysis method (out of class note)



* cation exchanger resins is used to separate. Free amino acid eluted.

This one is c terminal

3) Enzymatic method:



This one is c-terminal

Protein

Definition: Proteins are nitrogenous, amorphous subistance which occur in protoplasm of all animal and plant cell. They are colloid and have no characteristic melting point, some are crystalline all proteins are optically active (levorotatory) proteins may be coagulated, precipitated irreversibly by heat or by strong inorganic acid or bases. Example; Amylase, lipase, Insulin, Actin, keratin etc.

The composition of protein varies with source:

Carbon - 46-55 %
Hydrogen- 6-9 %
oxygen - 12-13 %
Nitrogen - 10-32%
Sulphur - 0.2-0.3%

No need

other element may also be present

Note: There is no sharp devide between peptides, polypeptides and proteins. one arbitrary convention designates proteins as those molecular weight above ~10000. Polypeptides are those molecular weight below ~10000. Protein and peptides differ in physical properti

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Denatured protein: All proteins are optically active, and may be coagulated and precipitated from aqueous solution by heat, the addition of acids, alkalis, salts, organic solvents miscible with water etc. Proteins in this precipitated state said to be denatured, and the process of reacing this state is called denaturation (Generally irreversible), occurs most readily near the isoelectric point. Associated with denaturation are changes in optical rotation, loss of biological activity (enzymes become inactive).

→ Ethanol, conc. solution of ammonium sulfate or sodium chloride can be used for denaturcation.

Extra: Denaturation is generally irreversible, but many examples are now known where the process has been reversed. The reversal process is called renaturation or refolding.

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classification of proteins:

3 types —

1) simple proteins: These give only one amino acids or their derivatives on hydrolysis.
   Example: Albumins, Globulins, prolamins.

2) conjgated proteins: These are proteins which contain a non protein group (i,e. a compound not containing amino acid residues) attached to the protein part. Ex. Nucleoproteins, Glycoproteins

3) Derived proteins: These are degradation products obtained by the action of acids, alkalis or enzymes on proteins.

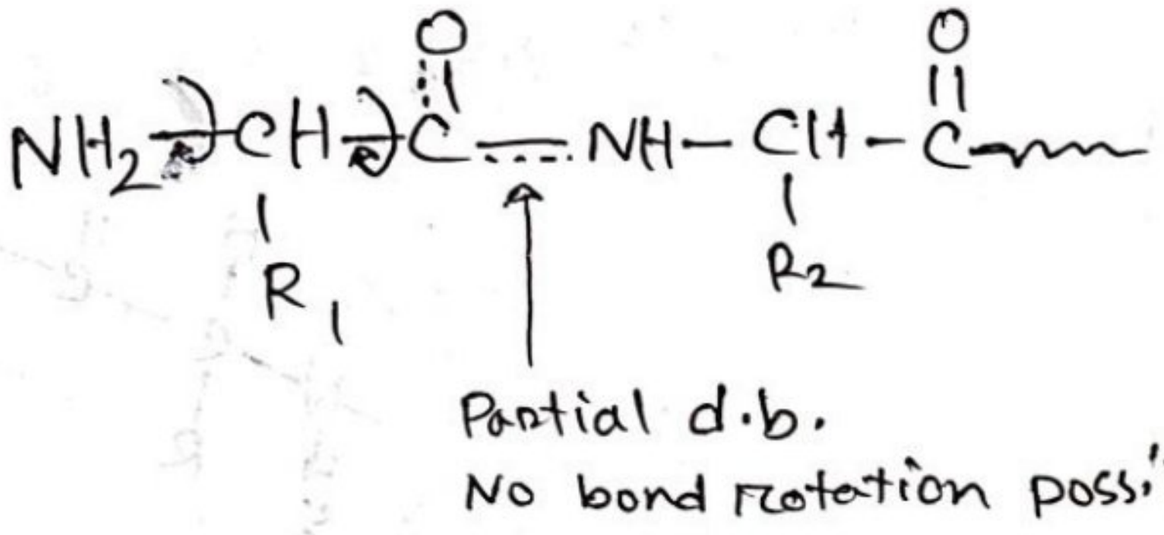
Structure of proteins:

1) Primary structure of proteins; Is the protein structure in which
→ multiple amino acids bound togethe in a sequential manner to form a polypeptide chain; in turn a number of plypeptid chains together forms a protein. Actually, proteins

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primary structure is the amino acid sequence in its polypeptide chain.

2) secondary structure: when the single bond rotates different shapes of peptide backbone (the backbone just refers to the polypeptide chain apart from R group atoms) or conformation can be established the structure is said to be secondary.



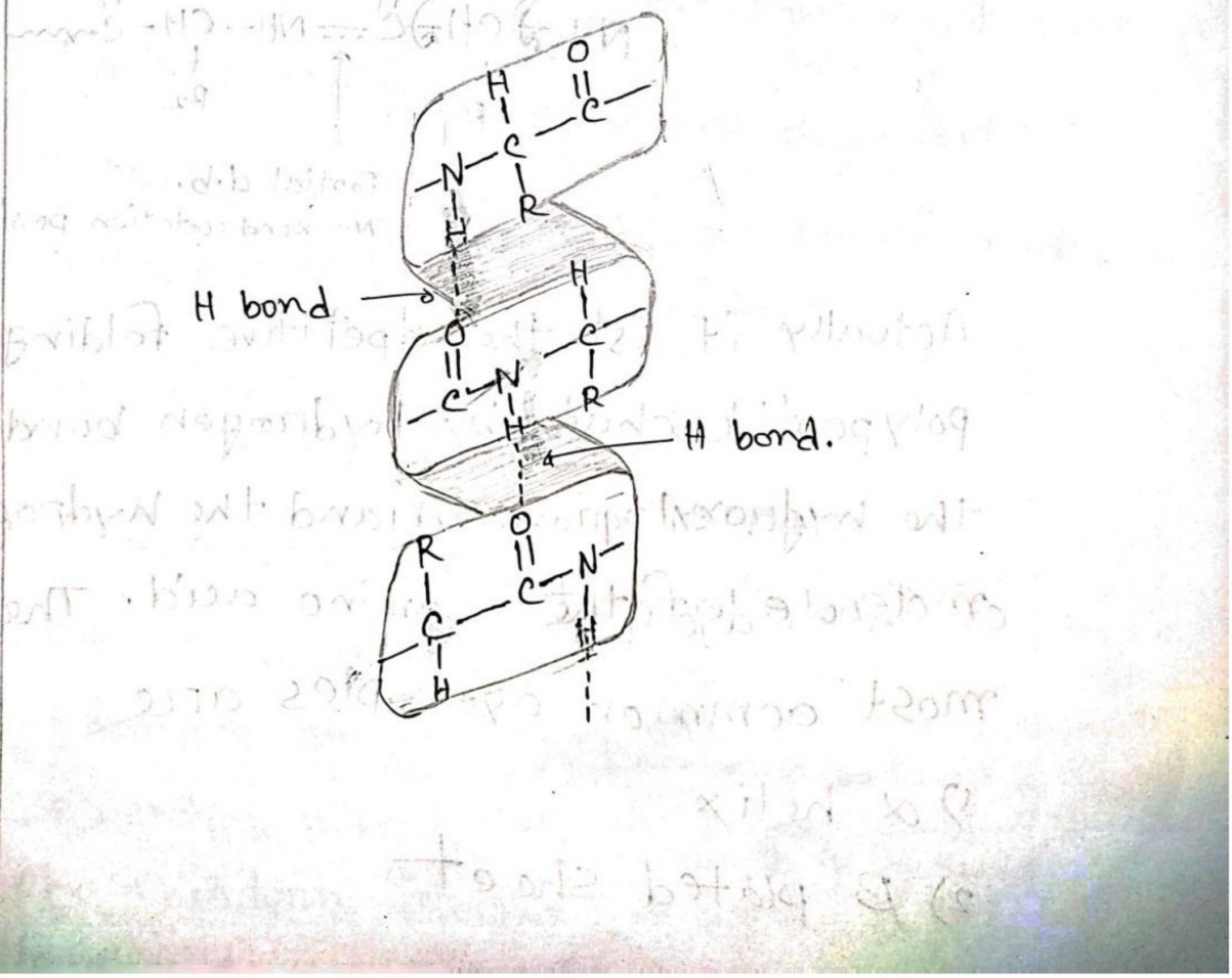
Actually it is the reptitive folding of polypeptide chain by hydrogen bond between the oxygen of peptide bond and the hydrogen of the adjacent amino acid. The most common examples are

1) α helix

2) β plated sheet

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α-helix: the carbonyl (-C=O) of one amino acid is hydrogen bonded to the amino hydrogen (N-H) of amino acid that is four down the chain. This pattern of bonding pulls the polypeptide chain into a helical structure that resembles a curled ribbon, with each turn of helix containing 3.6 amino acids.



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β-Plated sheet: two or more segments of a polypeptide chain line up next to each other, forming sheet like structure held together by hydrogen bonds. The hydrogen bonds form between carbonyl (>c=o) and amino groups of backbones while R groups extend above and below the plane of sheet. If R group is very large the β pated sheet is not formed.

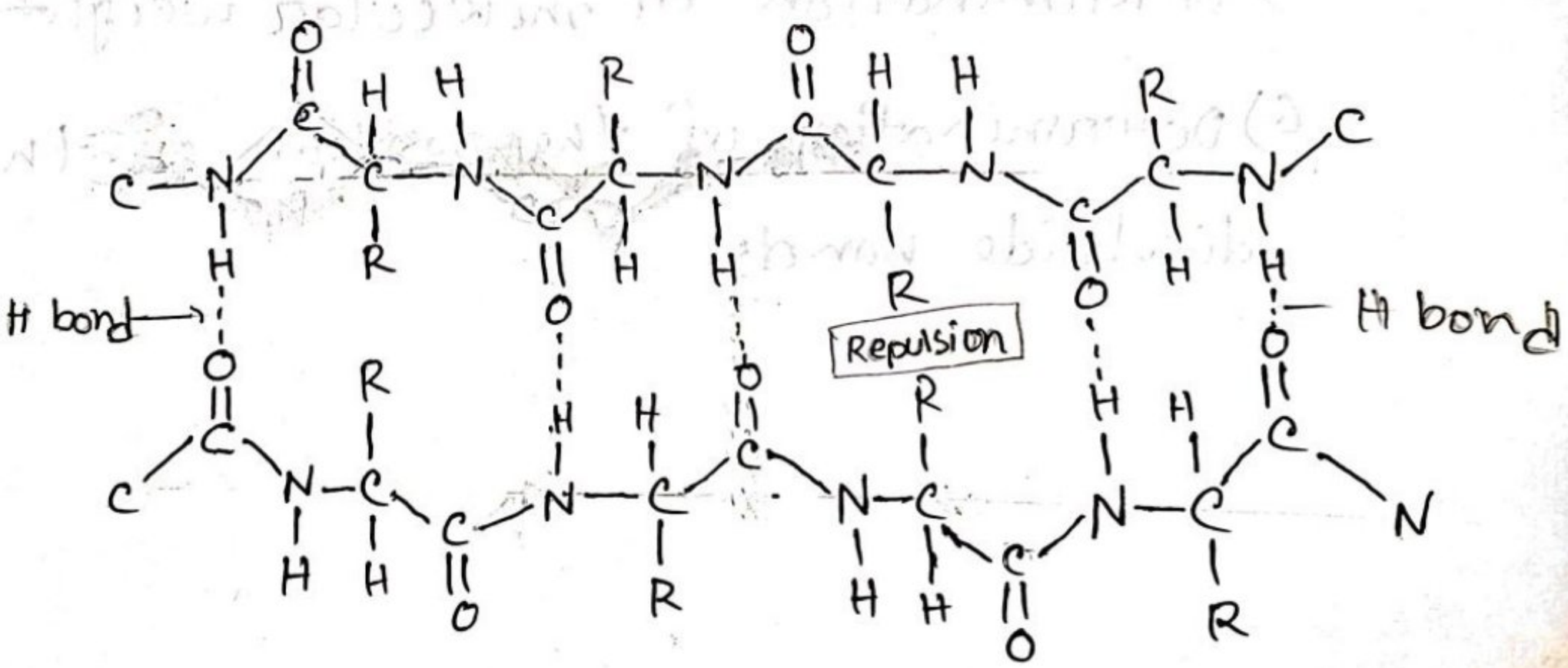


Fig. β plated sheet structure of protein.

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Q. How will you determine the structure of a primary proteins?

Ans: The steps are following

1) The peptide must be isolated and purified using Chromatographic method.

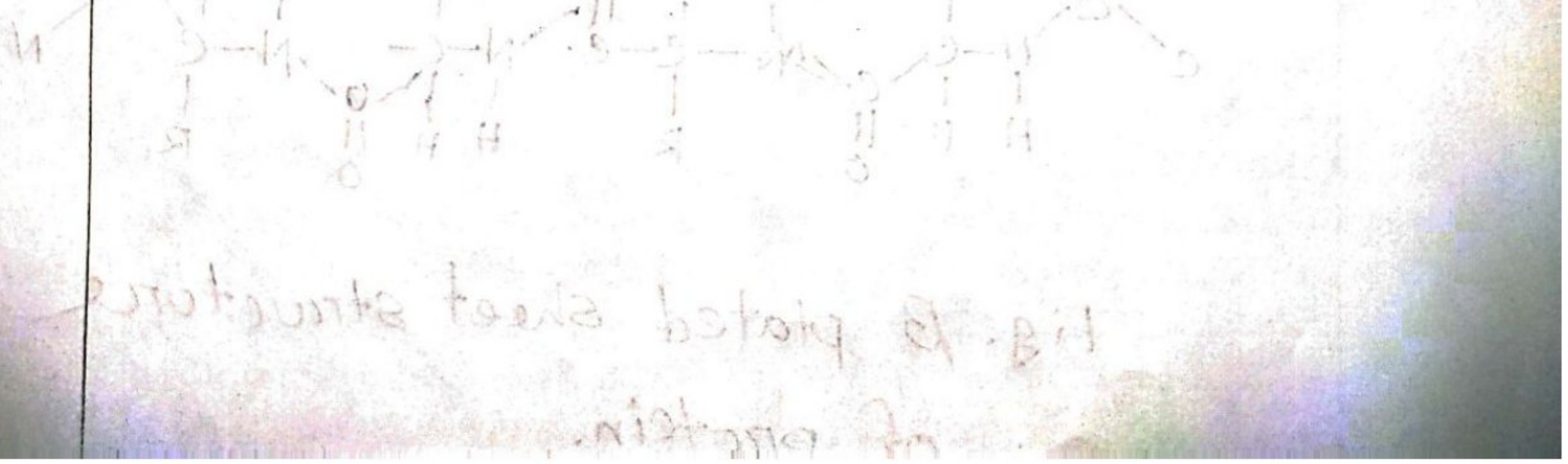
2) Determination of amino acid composition (by hydrolysis)

3) Determination of sequence of amino acid (may be by Edman method).

4) Determination of N and C terminal amino acids)

5) Determination of molecular weight

6) Determination of the position of the disulfide bonds.



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